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د. موسى رحمه جادالله عجال
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الفصل الأول 481
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Chapter 3

The Logarithmic Functions and Exponential Functions

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1- The natural logarithmic function

2- The natural exponential function

3- The general exponential functions and the general logarithmic functions

4- The integration of invers trigonometric functions

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1- The natural logarithmic function

The derivative of $\ln|x|$:

1. $\dfrac{d}{dx}\ln|x| = \dfrac{1}{x}$.
2. $\dfrac{d}{dx}\ln|f(x)| = \dfrac{f'(x)}{f(x)}$.

Note : $\ln|x|$ is the antiderivative of $\dfrac{1}{x}$.

Integration :

1. $\displaystyle\int \frac{1}{x}\,dx = \ln|x| + c$.
2. $\displaystyle\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$.

Some properties of $\ln|x|$: If $x, y > 0$ and $r \in \mathbb{R}$ then

1. $\ln(xy) = \ln x + \ln y$.
2. $\ln\left(\dfrac{x}{y}\right) = \ln x - \ln y$.
3. $\ln x^r = r \ln x$.

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Q: Find the derivatives $f'(x)$ of the following functions

$Q1:\ f(x)=\sqrt{x}\ln|x|$

Solution :

$$f'(x)=\sqrt{x}\times\frac{1}{x}+\ln|x|\times\frac{1}{2}x^{-\frac{1}{2}}$$

$$=\frac{1}{x}\sqrt{x}+\frac{1}{2}\frac{\ln|x|}{\sqrt{x}}$$

$Q2:\ y=\ln|x^3-4x^5+3x-1|$

Solution :

$$\frac{dy}{dx}=\frac{3x^2-20x^4+3}{x^3-4x^5+3x-1}$$

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(1) $f(x) = \ln\left|x^4 + x^3 + 1\right|$

(2) $f(x) = \ln\left|x^2 + \cos 2x\right|$

(3) $f(x) = \sin x \ \ln\left|5x\right|$

(4) $f(x) = \tan\left(\ln\left|3x\right|\right)$

(5) $f(x) = \left[3x + \ln\left|\sin x\right|\right]^8$

(6) $f(x) = \ln\left|\dfrac{\sqrt{x^2+1}\,\sin^5 x}{(x^3+4)^2}\right|$

(7) $f(x) = \dfrac{(3x+1)^{\frac{3}{2}}\,(x^2-1)^{\frac{2}{3}}}{\sqrt[3]{x^2+2}}$

(8) $f(x) = (x^2+1)^x$

(9) $f(x) = x^{\sin x}$

(10) $f(x) = (\cos x)^{x^3-1}$

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$$(3)\ f\ (x)=\sin(x)\ \ln|5x|$$

Solution :

$$f\ '(x)=\sin(x)\ \frac{1}{5x}+\ln|5x|\cos(x)$$

$$Q:f\ (x)=\left(\sin(x)\right)^{x^3}$$

Solution :

$$\ln|f\ (x)|=\ln\left|\left(\sin(x)\right)^{x^3}\right|$$

$$\ln|f\ (x)|=x^3\ln|\sin(x)|$$

$$\frac{f\ '(x)}{f\ (x)}=x^3\frac{\cos(x)}{\sin(x)}+\ln|\sin(x)|(3x^2)$$

$$f\ '(x)=\left[x^3\frac{\cos(x)}{\sin(x)}+\ln|\sin(x)|(3x^2)\right]f\ (x)$$

$$f\ '(x)=\left[x^3\cot(x)+3x^2\ln|\sin(x)|\right]\left(\sin(x)\right)^{x^3}$$

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$$(10)\ f\ (x)=\left(\cos(x)\right)^{x^3-1}$$

Solution :

$$\ln\left|f\ (x)\right|=\left(x^3-1\right)\ln\left|\cos(x)\right|$$

$$\frac{f\ '(x)}{f\ (x)}=\left(x^3-1\right)\frac{-\sin(x)}{\cos(x)}+\ln\left|\cos(x)\right|(3x^2)$$

$$f\ '(x)=\left[-\left(x^3-1\right)\tan(x)+3x^2\ln\left|\cos(x)\right|\right]f\ (x)$$

$$f\ '(x)=\left[-\left(x^3-1\right)\tan(x)+3x^2\ln\left|\cos(x)\right|\right]\left(\cos(x)\right)^{x^3-1}$$

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The integral natural logarithmic function

$$\int \frac{1}{x}\,dx = \ln|x| + c$$

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$$

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Q: Evaluate the following integrals

(1) $\int \frac{x^3+x+1}{x^4+2x^2+4x}\,dx$

(2) $\int \frac{x^2-\sin x}{x^3+3\cos x}\,dx$

(3) $\int \frac{x^2+3x-4}{x}\,dx$

(4) $\int \frac{\cos\left(\ln|x|\right)}{x}\,dx$

(5) $\int \frac{\left(8+\ln|x|\right)^5}{2x}\,dx$

(6) $\int \frac{1}{x\sqrt[3]{\ln|x|}}\,dx$

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$$Q1: \int \frac{x^2+1}{x^3+3x+7}dx$$

Solution :

$$\int \frac{x^2+1}{x^3+3x+7}dx = \frac{1}{3}\int \frac{3x^2+3}{x^3+3x+7}dx$$

$$= \frac{1}{3}\ln\left|x^3+3x+7\right|+c$$

$$Q2: \int \frac{x+\cos(x)}{x^2+2\sin(x)}dx$$

Solution :

$$\int \frac{x+\cos(x)}{x^2+2\sin(x)}dx = \frac{1}{2}\int \frac{2x+2\cos(x)}{x^2+2\sin(x)}dx$$

$$= \frac{1}{2}\ln\left|x^2+2\sin(x)\right|+c$$

$$Q3: \int \frac{1}{x\ln(x)}dx$$

Solution :

$$\int \frac{1}{x\ln(x)}dx = \int \frac{\frac{1}{x}}{\ln(x)}dx = \ln\left|\ln(x)\right|+c$$

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$$Q4: \int \frac{x^2+3x-4}{x}dx$$

Solution :

$$\int \frac{x^2+3x-4}{x}dx = \int \left(\frac{x^2}{x}+\frac{3x}{x}-\frac{4}{x}\right)dx$$

$$\int \left(x+3-4\frac{1}{x}\right)dx = \frac{1}{2}x^2+3x-4\ln|x|+c$$

$$Q5: \int x\sec(x^2-3)dx$$

Solution :

$$\int x\sec(x^2-3)dx = \frac{1}{2}\int 2x\sec(x^2-3)dx$$

$$= \frac{1}{2}\ln\left|\sec(x^2-3)+\tan(x^2-3)\right|+c$$

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2- The natural exponential function

Notes :

1. The domain of the function e^x is $\mathbb{R}$.
2. The range of the function e^x is the open interval $(0, \infty)$
3. $e^x > 0$ for every $x \in \mathbb{R}$.
4. $e^0 = 1$.
5. $e \approx 2.71828$ and $\ln(e) = 1$.
6. $\lim\limits_{x \to \infty} e^x = \infty$.
7. $\lim\limits_{x \to -\infty} e^x = 0$.
8. $\ln(e^x) = x$ and $e^{\ln x} = x$.

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Q: Find the derivatives $f'(x)$ of the following functions

$$\frac{d}{dx}\, e^{x} = e^{x}$$

$$\frac{d}{dx}\, e^{f(x)} = e^{f(x)}\, f'(x)$$

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(1) $f(x) = e^{x^3 - x + 2}$

(2) $f(x) = \dfrac{1}{e^{2x}} + e^{\cos 2x}$

(3) $f(x) = \left(\ln|3x| + e^{4x+2}\right)^8$

(4) $f(x) = \ln\left|e^{2\tan x} + \sec 3x\right|$

(5) $f(x) = e^{x^2} \ln\left|x^3 - 1\right|$

(6) $f(x) = \tan\left(e^{x^2+1}\right)$

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$$Q1: f\ (x) = e^{2x^5+3x}$$

Solution :

$$f'(x) = e^{2x^5+3x}\left(10x^4+3\right)$$

$$Q2: f\ (x) = e^{\sin(4x^4)} + \frac{1}{e^{-4x}}$$

Solution :

$$f\ (x) = e^{\sin(4x^4)} + e^{4x}$$

$$f'(x) = e^{\sin(4x^4)}\cos(4x^4)16x^3 + e^{4x}(4)$$

$$Q3: f\ (x) = \ln\left|e^{\tan(x)} + 3x^3 - 1\right|$$

Solution :

$$f'(x) = \frac{e^{\tan(x)}(\sec^2(x)) + 9x^2}{e^{\tan(x)} + 3x^3 - 1}$$

$$Q4: f\ (x) = \left(e^{3x} + 2x^4\right)^5$$

Solution :

$$f'(x) = 5\left(e^{3x} + 2x^4\right)^4\left(e^{3x}(3) + 8x^3\right)$$

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Q: Find $f'(x)$ of the following functions:

$$(1)\ f(x)=x^{3x^2}$$

$$(2)\ f(x)=(1+3x)^{\frac{1}{x}}$$

$$(3)\ f(x)=9^{\sqrt[3]{x}}$$

Solution:

$$(1)\ f(x)=x^{3x^2}$$

Solution:

$$\ln f(x)=\ln x^{3x^2}$$

$$\Rightarrow \ln f(x)=3x^2\ln x$$

$$\Rightarrow \frac{f'(x)}{f(x)}=3x^2.\frac{1}{x}+6x.\ln x$$

$$f'(x)=(3x+6x.\ln x)f(x)$$

$$f'(x)=(3x+6x.\ln x)x^{3x^2}$$

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The integration of natural exponential function

1- Integration Using "ln"

2- Integration Using "exp"

Theorem

Using the last properties of the logarithmic and exponential functions, we have

1. $\int \frac{dx}{x} = \ln|x| + c,$
2. $\int e^x dx = e^x + c,$
3. $\int \frac{u'(x)}{u(x)} dx = \ln|u(x)| + c,$
4. $\int u'(x)e^{u(x)} dx = e^{u(x)} + c.$

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(1) **Evaluate** $\int e^{8x+1}dx$

Solution :

$$\int e^{8x+1}dx = \frac{1}{8}e^{8x+1}+c$$

(2) **Evaluate** $\int xe^{x^2-3}dx$

Solution :

$$\int xe^{x^2-3}dx = \frac{1}{2}\int 2x\, e^{x^2-3}dx$$

$$= \frac{1}{2}e^{x^2-3}+c$$

(3) **Evaluate** $\int e^x \cos\left(e^x\right)dx$

Solution :

$$\int e^x \cos\left(e^x\right)dx = \sin\left(e^x\right)+c$$

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(4) **Evaluate** $\int \frac{e^{4x}}{\left(e^{4x}+3\right)^3}dx$

Solution :

$$\int \frac{e^{4x}}{\left(e^{4x}+3\right)^3}dx = \int e^{4x}\left(e^{4x}+3\right)^{-3}dx$$

$$\int e^{4x}\left(e^{4x}+3\right)^{-3}dx = \frac{1}{4}\int 4e^{4x}\left(e^{4x}+3\right)^{-3}dx$$

$$\frac{1}{4}\frac{\left(e^{4x}+3\right)^{-2}}{-2}+c = -\frac{1}{8}\left(e^{4x}+3\right)^{-2}+c$$

(5) **Evaluate** $\int \frac{e^{8\ln x}}{x^4}dx$

Solution :

$$\int \frac{e^{8\ln x}}{x^4}dx = \int \frac{e^{\ln x^8}}{x^4}dx = \int \frac{x^8}{x^4}dx$$

$$\int x^4 dx = \frac{1}{5}x^5 + c$$

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(12) **Evaluate** $\int e^{7\ln|x+1|}dx$

Solution:

$$\int e^{7\ln|x+1|}dx = \int e^{\ln|x+1|^7}dx$$

$$\int (x+1)^7 dx = \frac{1}{8}(x+1)^8 + c$$

(5) **Evaluate** $\int \frac{e^{\cos(3x)}}{\csc(3x)}dx$

Solution:

$$\int \frac{e^{\cos(3x)}}{\csc(3x)} = \int \sin(3x)e^{\cos(3x)}dx$$

$$= -\frac{1}{3}\int -3\sin(3x)e^{\cos(3x)}dx$$

$$= -\frac{1}{3}e^{\cos(3x)} + c$$

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Evaluate the following integrals

(1) $\int 5e^{8x+1}\,dx$

(2) $\int 7x^2\,e^{x^3-4}\,dx$

(3) $\int \frac{e^{\frac{1}{x}}}{x^2}\,dx$

(4) $\int \frac{e^{\sqrt[3]{x}}}{x^{\frac{2}{3}}}\,dx$

(5) $\int \frac{e^{\cos 3x}}{\csc 3x}\,dx$

(6) $\int \frac{e^{\cot 2x}}{\sin^2 2x}\,dx$

(7) $\int_{\ln 2}^{\ln 5} 3e^x\,dx$

(8) $\int e^{7x}\,\csc^2\left(2+e^{7x}\right)\,dx$

(9) $\int \frac{e^{3x}}{e^{3x}+2}\,dx$

(10) $\int \frac{e^{3x}}{\left(e^{3x}+2\right)^5}\,dx$

(11) $\int e^x\left(4-\cos\left(e^x\right)\right)\,dx$

(12) $\int e^{7\ln|x+1|}\,dx$

(13) $\int \frac{e^{2\ln|x|}}{x^3}\,dx$

(14) $\int e^{3x^2+\ln|x|}\,dx$

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3- The general exponential functions and the general logarithmic functions

إشتقاق الدالة الأسية العامة :

$$\frac{d}{dx}\, a^x = a^x \ln a$$

$$\frac{d}{dx}\, a^{f(x)} = a^{f(x)} f'(x)\ \ln a$$

إشتقاق الدالة اللوغاريتمية العامة :

$$\frac{d}{dx}\, \log_a |x| = \frac{1}{\ln a}\,\frac{1}{x} = \frac{1}{x \ln a}$$

$$\frac{d}{dx}\, \log_a |f(x)| = \frac{1}{\ln a}\,\frac{f'(x)}{f(x)} = \frac{f'(x)}{f(x)\ \ln a}$$

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General Notes:

$$\ln x = \log_e x \text{ و } \log x = \log_{10} x \quad (1)$$

$$\log_a x = y \iff x = a^y \quad (2)$$

$$\log_a a = 1 \quad (3)$$

$$\log_a x = \frac{\ln x}{\ln a} \quad (4)$$

$$\log_a (xy) = \log_a x + \log_a y \quad (1)$$

$$\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y \quad (2)$$

$$\log_a x^r = r \log_a x \quad (3)$$

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Find the derivatives of the following functions

$(1)\ f(x)=5^{3x^2-2x}$

Solution :

$$f'(x)=5^{3x^2-2x}\,(6x-2)\ln|3|$$

$(2)\ f(x)=\pi^{\frac{3}{x}}$

Solution :

$$f'(x)=\pi^{\frac{3}{x}}\left(-3x^{-2}\right)\ln\pi$$

$(3)\ f(x)=\ln\left|4^{x^2}+3x^5\right|$

Solution :

$$f'(x)=\frac{4^{x^2}(2x)\ln 4+15x^4}{4^{x^2}+3x^5}$$

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Derivatives of General Exponential Functions

$$(1)\ \frac{d}{dx}(a^x) = a^x \ln(a)$$

$$(2)\ \frac{d}{dx}(a^u) = a^u \ln(a)\frac{du}{dx}$$

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Find the derivative of the following functions:

(1) $f(x) = 7^x$

(2) $f(x) = 8^{\sqrt[3]{x}}$

(3) $f(x) = (5^x + 5^{-x})^5$

(4) $f(x) = (4^{x^2} + 7^{3x+1})^6$

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(1) $f(x)=7^{x}$

Solution :

$f'(x)=7^{x}\ln 7$

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(4) $f(x)=(4^{x^2}+7^{3x+1})^{6}$

Solution :

$f'(x)=6(4^{x^2}+7^{3x+1})^{5}\cdot\left[4^{x^2}\cdot\ln 4\cdot(2x)+7^{3x+1}\cdot\ln 7\cdot(3)\right]$

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$(1)\ f(x)=\log_5\left|2x^5+\sin(4x)\right|$

Solution:

$$f'(x)=\frac{1}{\ln 5}\frac{10x^4+4\cos(4x)}{2x^5+\sin(4x)}$$

$(2)\ f(x)=\sin\left[\log_8\left|2x+5\right|\right]$

Solution:

$$f'(x)=\cos\left[\log_8\left|2x+5\right|\right]\frac{1}{\ln 8}\frac{2}{2x+5}$$

$(3)\ f(x)=\tan(3x^3)\log_3\left|x+1\right|$

Solution:

$$f(x)=\tan(3x^3)\frac{1}{\ln 3}\frac{1}{x+1}+\log_3\left|x+1\right|\left(\sec^2(3x^3)\right)(9x^2)$$

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Find the derivatives of the following functions

$$(1)\ f(x) = 5^{\sin 2x}$$

$$(3)\ f(x) = \ln\left|3^{\cos x} + e^{5x}\right|$$

$$(5)\ f(x) = \log_5\left|x^4 - \csc 6x\right|$$

$$(7)\ f(x) = \sqrt{x}\ \log_6\left|2 + \cot x\right|$$

$$(2)\ f(x) = \tan\left(7^{x^2}\right)$$

$$(4)\ f(x) = \left(\sec x^2 + \pi^{2x-1}\right)^5$$

$$(6)\ f(x) = \sin\left(\log_3\left|x^2 - |1|\right|\right)$$

$$(8)\ f(x) = \left(\log_2|3x-2| + \sqrt[4]{x}\right)^8$$

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Integrations of general exponential and logarithmic functions

تكامل الدالة الأسية العامة :

$$\int a^x \, dx = \frac{a^x}{\ln a} + c$$

$$\int a^{f(x)} \, f'(x) \, dx = \frac{a^{f(x)}}{\ln a} + c$$

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(1) **Evaluate** $\int 7^x \left(7^x + 4\right)^{11} dx$

Solution :

$$\frac{1}{\ln 7}\int \left(7^x \ln 7\right)\left(7^x + 4\right)^{11} dx$$

$$= \frac{1}{\ln 7}\frac{\left(7^x + 4\right)^{12}}{12} + c$$

(2) **Evaluate** $\int 3^x \left(1 + \cos(3^x)\right) dx$

Solution :

$$= \int \left(3^x + 3^x \cos(3^x)\right) dx$$

$$= \int \left(3^x\right) dx + \frac{1}{\ln 3}\int \left(3^x \ln(3) \cos(3^x)\right) dx$$

$$= \frac{3^x}{\ln 3} + \frac{1}{\ln 3}\sin(3^x) + c$$

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Find the integration of the following functions:

(1) $\int 7^x \, dx$ (2) $\int \frac{(1+4^x)^2}{4^x} dx$

(3) $\int \frac{3^{\cot(x)}}{\sin^2(x)} dx$ (4) $\int 4^x \, 5^{4^x} \, dx$

(1) $\int 7^x \, dx$

Solution :

$$\int 7^x \, dx = \frac{7^x}{\ln 7} + c$$

---

(3) $\int \frac{3^{\cot(x)}}{\sin^2(x)} dx$

Solution :

$$u = \cot x \Rightarrow du = -\csc^2 x \, dx$$

$$= -\int 3^u (-\csc^2 x) dx$$

$$= -\int 3^u du = \frac{3^u}{\ln 3} + c = \frac{3^{\cot x}}{\ln 3} + c$$

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$$(4)\ \int 4^x\, 5^{4^x}\, dx$$

Solution :

$$u = 4^x \Rightarrow du = 4^x \cdot \ln 4\, dx$$

$$\Rightarrow \frac{1}{\ln 4} du = 4^x\, dx$$

$$\int 5^u \frac{1}{\ln 4} du = \frac{1}{\ln 4} \int 5^u\, du$$

$$= \frac{1}{\ln 4} \cdot \frac{5^u}{\ln 5} + c = \frac{1}{\ln 4} \cdot \frac{5^{4^x}}{\ln 5} + c$$

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$$\int x^2 6^{x^3-2}\,dx \quad (1)$$

$$\int x^2 6^{x^3-2}\,dx = \frac{1}{3}\int 6^{x^3-2}(3x^2)\,dx = \frac{1}{3}\,\frac{6^{x^3-2}}{\ln 6} + c$$

$$\int \frac{3^{\cot x}}{\sin^2 x}\,dx \quad (2)$$

$$\int \frac{3^{\cot x}}{\sin^2 x}\,dx = \int 3^{\cot x}\frac{1}{\sin^2 x}\,dx = \int 3^{\cot x}\csc^2 x\,dx$$

$$= -\int 3^{\cot x}(-\csc^2 x)\,dx = -\frac{3^{\cot x}}{\ln 3} + c$$

$$\int \left(5^x + \frac{1}{2^x}\right)\,dx \quad (3)$$

$$\int \left(5^x + \frac{1}{2^x}\right)\,dx = \int (5^x + 2^{-x})\,dx = \int 5^x\,dx + \int 2^{-x}\,dx$$

$$= \int 5^x\,dx + \frac{1}{-1}\int 2^{-x}(-1)\,dx = \frac{5^x}{\ln 5} - \frac{2^{-x}}{\ln 2} + c$$

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Evaluate the following integrals

(1) $\int \frac{5\sqrt{x}}{\sqrt{x}}\,dx$

(2) $\int \frac{\pi^{\sin 3x}}{\sec 3x}\,dx$

(3) $\int x3^{x^2+1}\cos\left(3^{x^2+1}\right)dx$

(4) $\int \frac{\sin x\, 7^{\cos x}}{7^{\cos x}+3}\,dx$

(5) $\int \frac{6^{3x}}{(6^{3x}+2)9}\,dx$

(6) $\int 2^x 5^{2^x}\,dx$

(7) $\int x\,4^{x^2}\left(1+\sec\left(4^{x^2}\right)\right)dx$

(8) $\int \frac{(4+\log_2 x)^7}{3x}\,dx$

(9) $\int \frac{\cos(\log 3x)}{2x}\,dx$

(10) $\int \frac{1}{x\log_3 x}\,dx$

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4- The integration of invers trigonometric functions

$$\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}}\quad,\quad |x|<1 \quad (1)$$

$$\frac{d}{dx}\sin^{-1}\left(f(x)\right)=\frac{f'(x)}{\sqrt{1-[f(x)]^2}}\quad,\quad |f(x)|<1$$

$$\frac{d}{dx}\cos^{-1}x=-\frac{1}{\sqrt{1-x^2}}\quad,\quad |x|<1 \quad (2)$$

$$\frac{d}{dx}\cos^{-1}\left(f(x)\right)=-\frac{f'(x)}{\sqrt{1-[f(x)]^2}}\quad,\quad |f(x)|<1$$

$$\frac{d}{dx}\tan^{-1}x=\frac{1}{1+x^2} \quad (3)$$

$$\frac{d}{dx}\tan^{-1}\left(f(x)\right)=\frac{f'(x)}{1+[f(x)]^2}$$

$$\frac{d}{dx}\cot^{-1}x=-\frac{1}{1+x^2} \quad (4)$$

$$\frac{d}{dx}\cot^{-1}\left(f(x)\right)=-\frac{f'(x)}{1+[f(x)]^2}$$

$$\frac{d}{dx}\sec^{-1}x=\frac{1}{x\sqrt{x^2-1}}\quad,\quad |x|>1 \quad (5)$$

$$\frac{d}{dx}\sec^{-1}\left(f(x)\right)=\frac{f'(x)}{f(x)\sqrt{[f(x)]^2-1}}\quad,\quad |f(x)|>1$$

$$\frac{d}{dx}\csc^{-1}x=-\frac{1}{x\sqrt{x^2-1}}\quad,\quad |x|>1 \quad (6)$$

$$\frac{d}{dx}\csc^{-1}\left(f(x)\right)=-\frac{f'(x)}{f(x)\sqrt{[f(x)]^2-1}}\quad,\quad |f(x)|>1$$

1. $\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}, \forall\ |x| < 1.$
2. $\frac{d}{dx}\cos^{-1}(x) = \frac{-1}{\sqrt{1-x^2}}, \forall\ |x| < 1.$
3. $\frac{d}{dx}\tan^{-1}(x) = \frac{1}{1+x^2}, \forall\ x \in \mathbb{R}.$
4. $\frac{d}{dx}\cot^{-1}(x) = \frac{-1}{1+x^2}, \forall\ x \in \mathbb{R}.$
5. $\frac{d}{dx}\sec^{-1}(x) = \frac{1}{|x|\sqrt{x^2-1}}, \forall\ |x| > 1.$
6. $\frac{d}{dx}\csc^{-1}(x) = \frac{-1}{|x|\sqrt{x^2-1}}, \forall\ |x| > 1.$

For $a > 0$,

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c \ , \ (|x| < a)$
2. $\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}dx = \sin^{-1}\left(\frac{f(x)}{a}\right) + c \ , \ (|f| < a))$
3. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
4. $\int \frac{f'(x)}{a^2+[f(x)]^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{f(x)}{a}\right) + c$
5. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c \ , \ (x > a)$
6. $\int \frac{f'(x)}{f(x)\sqrt{[f(x)]^2-a^2}}dx = \frac{1}{a}\sec^{-1}\left(\frac{f(x)}{a}\right) + c \ , \ (f > a))$

Q1: Find the derivatives of the each of the following functions

$$(a)\ f(x) = 2\sinh^{-1}(x-1)$$

$$(b)\ f(x) = \sec^{-1}(2x)$$

$$(c)\ f(x) = \tanh^{-1}(\sin(2x^5))$$

$$(d)\ f(x) = \tan^{-1}x + x\cot^{-1}x$$

$$(e)\ f(x) = \ln\left(\cosh^{-1}(3x^2-5)\right)$$

$$(k)\ f(x) = 4^{\cosh^{-1}(4x^2)}$$

1 $f(x)=e^{\cos^{-1}(4x^2+1)}$

Solution :

$$f'(x)=e^{\cos^{-1}(4x^2+1)}\frac{-1}{\sqrt{1-(4x^2+1)^2}}(8x)$$

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2 $f(x)=\left[5^x+\tan^{-1}(2x+1)\right]^5$

Solution :

$$f'(x)=5\left[5^x+\tan^{-1}(2x+1)\right]^4\left[5^x\ln 5+\frac{1}{1+(2x+1)^2}(2)\right]$$

$$(c)\ f(x)=\tanh^{-1}(\sin(2x^5))$$

Solution :

$$f'(x)=\frac{1}{1-(\sin(2x^5))^2}\cdot\cos(2x^5)\cdot(10x^4)$$

$$f'(x)=\frac{10x^4\cos(2x^5)}{1-(\sin(2x^5))^2}$$

$$(e)\ f(x)=\ln\left(\cosh^{-1}(3x^2-5)\right)$$

Solution :

$$f'(x)=\frac{1}{\cosh^{-1}(3x^2-5)}\frac{1}{\sqrt{(3x^2-5)^2-1}}(6x)$$

$$(k)\ f(x)=4^{\cosh^{-1}(4x^2)}$$

Solution :

$$f'(x)=\ln(4)\cdot 4^{\cosh^{-1}(4x^2)}\cdot\frac{1}{\sqrt{(4x^2)^2-1}}\cdot(8x)$$

Q2: Find the derivatives of the each of the following functions

$$(1)\ f(x) = \sqrt{x}\ \sin^{-1}(5x)$$

$$(3)\ f(x) = \ln\left|x^3 + \tan^{-1}(2x)\right|$$

$$(5)\ f(x) = e^{\sec^{-1}(4x)}$$

$$(2)\ f(x) = \ln|3x-1|\ \cos^{-1}\left(\sqrt{x}\right)$$

$$(4)\ f(x) = \cot^{-1}\left(e^{3x}\right) + 5^{\sqrt[3]{x}}$$

$$(6)\ f(x) = \left(\csc^{-1}\left(\frac{1}{x}\right) + x^4\right)^7$$

Q: Evaluate the following integrals

1 **Evaluate** $\int \frac{x}{\sqrt{16-x^4}}dx$

Solution:

$$\int \frac{x}{\sqrt{4^2-(x^2)^2}}dx = \frac{1}{2}\int \frac{2x}{\sqrt{4^2-(x^2)^2}}dx$$

$$= \frac{1}{2}\sin^{-1}\left(\frac{x^2}{4}\right)+c$$

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2 **Evaluate** $\int \frac{x^2}{9+x^6}dx$

Solution:

$$\int \frac{x^2}{3^2+(x^3)^2}dx = \frac{1}{3}\int \frac{3x^2}{3^2+(x^3)^2}dx$$

$$\frac{1}{3}\times\frac{1}{3}\int \frac{3x^2}{3^2+(x^3)^2}dx = \frac{1}{9}\tan^{-1}\left(\frac{x^3}{3}\right)+c$$

3 **Evaluate** $\int \frac{e^{2x}}{25+e^{4x}}dx$

Solution :

$$\int \frac{e^{2x}}{25+e^{4x}}dx = \int \frac{e^{2x}}{5^2+\left(e^{2x}\right)^2}dx$$

$$\frac{1}{2}\int \frac{2e^{2x}}{5^2+\left(e^{2x}\right)^2}dx = \frac{1}{2}\times\frac{1}{5}\tan^{-1}\left(\frac{e^{2x}}{5}\right)+c$$

$$= \frac{1}{10}\tan^{-1}\left(\frac{e^{2x}}{5}\right)+c$$

4 **Evaluate** $\int \frac{1}{x\sqrt{x^4-16}}dx$

Solution :

$$\int \frac{1}{x\sqrt{x^4-16}}dx = \frac{1}{2}\int \frac{2x}{x^2\sqrt{(x^2)^2-4^2}}dx$$

$$\frac{1}{2}\times\frac{1}{4}\sec^{-1}\left(\frac{x^2}{4}\right)+c = \frac{1}{8}\sec^{-1}\left(\frac{x^2}{4}\right)+c$$

5 **Evaluate** $\int \frac{\sin x}{\sqrt{25-\cos^2 x}}dx$

Solution :

$$\int \frac{\sin x}{\sqrt{25-\cos^2 x}}dx = \int \frac{\sin x}{\sqrt{5^2-(\cos x)^2}}dx$$

$$-\int \frac{-\sin x}{\sqrt{5^2-(\cos x)^2}}dx = -\sin^{-1}\left(\frac{\cos x}{5}\right)+c$$

ملاحظة: يمكننا أن نستخدم طريقة أخرى وهي التكامل بالتعويض

(1) $\int \frac{x}{\sqrt{1-x^4}}dx$

Solution :

$$\int \frac{x}{\sqrt{1-(x^2)^2}}dx$$

$$Let\ \ u=x^2 \Rightarrow du=2x\,dx \Rightarrow \frac{1}{2}du=x\,dx$$

$$\int \frac{1}{\sqrt{1-u^2}}\frac{1}{2}du \Rightarrow \frac{1}{2}\int \frac{1}{\sqrt{1-u^2}}du=\frac{1}{2}\sin^{-1}u+c$$

$$=\frac{1}{2}\sin^{-1}(x^2)+c$$

Q4: Evaluate the integrals

(1) $\int \frac{x}{\sqrt{25-9x^4}}\,dx$

(2) $\int \frac{5x^2}{x^6+36}\,dx$

(3) $\int \frac{1}{x\sqrt{x^8-16}}\,dx$

(4) $\int \frac{3x}{\sqrt{4-(x^2+1)^2}}\,dx$

(5) $\int \frac{\cos x}{1+\sin^2 x}\,dx$

(6) $\int \frac{7}{25+(x-1)^2}\,dx$

(7) $\int \frac{1}{\sqrt{e^{4x}-16}}\,dx$

(8) $\int \frac{1}{x\sqrt{x^3-36}}\,dx$

(9) $\int \frac{2}{\sqrt{-x^2+2x+8}}\,dx$

(10) $\int \frac{3}{x^2+4x+13}\,dx$

(11) $\int \frac{5}{(x-2)\sqrt{x^2-4x}}\,dx$

(12) $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}}\,dx$

(13) $\int \frac{x-4}{\sqrt{9-x^2}}\,dx$

(14) $\int \frac{x+5}{x^2+36}\,dx$

(15) $\int \frac{x+\tan^{-1} x}{1+x^2}\,dx$

(16) $\int \frac{3-\sin^{-1} x}{\sqrt{1-x^2}}\,dx$

10 **Evaluate** $\int \frac{3}{x^2+4x+13}dx$

Solution:

$$x^2+4x+13 \Rightarrow \left(\frac{4}{2}\right)^2 = 4$$

$$\Rightarrow x^2+4x+13+4-4$$

$$\Rightarrow x^2+4x+4+13-4$$

$$\Rightarrow x^2+4x+4+9$$

$$\Rightarrow x+2 \quad x+2 \quad +3^2$$

$$\Rightarrow x+2^2+3^2 \Rightarrow \int \frac{3}{x+2^2+3^2}dx$$

$$\Rightarrow 3\int \frac{1}{x+2^2+3^2}dx$$

$$= \int \frac{3}{x^2+4x+13}dx$$

$$= \frac{3}{3}\tan^{-1}\left(\frac{x+2}{3}\right)+c$$

$$= \tan^{-1}\left(\frac{x+2}{3}\right)+c$$

د. موسى رحمه جادالله عجال
العام الجامعي 1449/1448هـ

Hyperbolic Functions

The Derivatives and Integration of Hyperbolic Functions

$$(1)\ \frac{d}{dx}\left(\sinh u\right)=\cosh u\cdot\frac{du}{dx} \qquad (2)\ \frac{d}{dx}\left(\cosh u\right)=\sinh u\cdot\frac{du}{dx}$$

$$(3)\ \frac{d}{dx}\left(\tanh u\right)=\operatorname{sech}^2 u\cdot\frac{du}{dx} \qquad (4)\ \frac{d}{dx}\left(\coth u\right)=-\operatorname{csch} u\cdot\frac{du}{dx}$$

$$(5)\ \frac{d}{dx}\left(\operatorname{sech} u\right)=-\operatorname{sech} u\ \tanh u\cdot\frac{du}{dx} \qquad (6)\ \frac{d}{dx}\left(\operatorname{csch} u\right)=-\operatorname{csch} u\ \coth u\cdot\frac{du}{dx}$$

$$(1)\int \sinh u\,du=\cosh u+c \qquad (2)\int \cosh u\,du=\sinh u+c$$

$$(3)\int \sinh^2 u\,du=-\tanh u+c \qquad (4)\int \operatorname{csch}^2 u\,du=-\coth u+c$$

$$(5)\int \operatorname{sech} u\tanh u\,du=-\operatorname{sech} u+c \qquad (6)\int \operatorname{csch} u\coth u\,du=-\operatorname{csch} u+c$$

$$(1)\ \sinh(x) = \frac{e^{x} - e^{-x}}{2}$$

$$(2)\ \cosh(x) = \frac{e^{x} + e^{-x}}{2}$$

$$(3)\ \tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

$$(4)\ \coth(x) = \frac{\cosh(x)}{\sinh(x)} = \frac{e^{x} + e^{-x}}{e^{x} - e^{-x}}$$

$$(5)\ \operatorname{sech}(x) = \frac{1}{\sinh(x)} = \frac{2}{e^{x} + e^{-x}}$$

$$(6)\ \operatorname{csch}(x) = \frac{1}{\cosh(x)} = \frac{2}{e^{x} - e^{-x}}$$

Q1: Find the derivatives of the following functions

(1) $f\ (x\ ) = e^{\sinh(3x^{\,5})}$

Solution :

$$f\ '(x\ ) = e^{\sinh(3x^{\,5})}\cosh(3x^{\,5})(15x^{\,4})$$

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(2) $f\ (x\ ) = \ln\left|\tanh\ \ 2-7x^{\,-2}\ \right|$

Solution :

$$f\ '(x\ ) = \frac{\operatorname{sech}^{2}\ \ 2-7x^{\,-2}\ \ \ 14x^{\,-1}}{\tanh\ \ 2-7x^{\,-2}}$$

(3) $f\ (x)=\cosh\ 5^{x+1}$

Solution:

$$f'(x)=\sinh\ 5^{x+1}\ 5^{x+1}\ln 5$$

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(4) $f\ (x)=\left[\operatorname{csch}\ 5x\ +1\right]^{8}$

Solution:

$$f'(x)=8\left[\operatorname{csch}\ 5x\ +1\right]^{7}\ -\operatorname{csch}\ 5x\ \coth\ 5x\ \ 5$$

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(5) $f\ (x)=x^{\sinh(x)}$

Solution:

$$\ln\left|f\ (x)\right|=\ln\left|x^{\sinh(x)}\right|$$

$$\ln\left|f\ (x)\right|=\sinh(x)\ln\left|x\right|$$

$$\frac{f'(x)}{f\ (x)}=\sinh(x)\frac{1}{x}+\ln\left|x\right|\cosh(x)$$

$$f'(x)=\left[\sinh(x)\frac{1}{x}+\ln\left|x\right|\cosh(x)\right]f\ (x)$$

$$f'(x)=\left[\sinh(x)\frac{1}{x}+\ln\left|x\right|\cosh(x)\right]\ln\left|x^{\sinh(x)}\right|$$

Q2: Find the derivatives of the following functions

$$(1)\ f(x) = \sinh\left(e^{2x}\right)$$

$$(3)\ f(x) = \tanh^{-1}\left(\sin^{-1} x\right)$$

$$(5)\ f(x) = csch\left(x^2\right)\ \cos 3x$$

$$(7)\ f(x) = \ln\left|cosh\left(x^2\right) - 4^x\right|$$

$$(2)\ f(x) = \cosh\left(\ln|x|\right)$$

$$(4)\ f(x) = sech\left(\frac{1}{x}\right)$$

$$(6)\ f(x) = \left(\tanh\sqrt{x} + \coth\sqrt[3]{x}\right)^6$$

$$(8)\ f(x) = e^{\sinh^2 x}$$

Q1: Evaluate the following integrals

(1) $\int \operatorname{sech}^2(x)\,dx$

Solution:

$$\int \operatorname{sech}^2(x)\,dx = -\coth(x) + c$$

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(2) $\int x^2 \cosh(x^3)\,dx$

Solution:

$$\int x^2 \cosh(x^3)\,dx = \frac{1}{3}\int 3x^2 \cosh(x^3)\,dx$$

$$= \frac{1}{3}\sinh(x^3) + c$$

(3) $\int \frac{\operatorname{sech}^2 \sqrt{x}}{\sqrt{x}} dx$

Solution :

$$\int \frac{\operatorname{sech}^2 \sqrt{x}}{\sqrt{x}} dx = 2\int \frac{1}{2\sqrt{x}} \operatorname{sech}^2 \sqrt{x}\ dx$$

$$= 2\tanh \sqrt{x} + c$$

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(4) $\int e^{\tanh(x)} \operatorname{sech}^2 x\ dx$

Solution :

$$\int e^{\tanh(x)} \operatorname{sech}^2 x\ dx = e^{\tanh(x)} + c$$

Q2: Evaluate the following integrals

(1) $\int e^{4x} \sinh\left(5 - e^{4x}\right) dx$

(2) $\int \frac{\cosh\left(\ln|x|\right)}{5x} dx$

(3) $\int \frac{sech^2\left(\frac{1}{x^2}\right)}{x^3} dx$

(4) $\int \frac{sech\sqrt{x}\,\tanh\sqrt{x}}{\sqrt{x}} dx$

(5) $\int \frac{\coth\left(\sin x\right)}{\sec x} dx$

(6) $\int \frac{e^{\cosh 3x}}{csch3x} dx$

(7) $\int \frac{5^{sechx} \sinh x}{\cosh^2 x} dx$

(8) $\int \frac{sech4x\,\tanh 4x}{2 + sech4x} dx$

(9) $\int \frac{csch^2 x}{4 + \coth^2 x} dx$

(10) $\int \frac{3}{cschx\sqrt{9 - \cosh^2 x}} dx$

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Inverse of Hyperbolic Functions

Inverse of Hyperbolic Functions

$$(1)\ \frac{d}{dx}\left(\sinh^{-1}u\right)=\frac{1}{\sqrt{u^2+1}}\cdot\frac{du}{dx}\quad\Rightarrow\quad\int\frac{1}{\sqrt{a^2+u^2}}du=\sinh^{-1}\left(\frac{u}{a}\right)+c$$

$$(2)\ \frac{d}{dx}\left(\cosh^{-1}u\right)=\frac{1}{\sqrt{u^2-1}}\cdot\frac{du}{dx}\quad\Rightarrow\quad\int\frac{1}{\sqrt{u^2-a^2}}du=\cosh^{-1}\left(\frac{u}{a}\right)+c$$

$$(3)\ \frac{d}{dx}\left(\tanh^{-1}u\right)=\frac{1}{1-u^2}\cdot\frac{du}{dx}\quad\Rightarrow\quad\int\frac{1}{a^2-u^2}du=\frac{1}{a}\tanh^{-1}\left(\frac{u}{a}\right)+c$$

$$(4)\ \frac{d}{dx}\left(\operatorname{sech}^{-1}u\right)=\frac{-1}{u\sqrt{1-u^2}}\cdot\frac{du}{dx}\quad\Rightarrow\quad\int\frac{1}{u\sqrt{a^2-u^2}}du=-\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{u}{a}\right)+c$$

$$(5)\ \frac{d}{dx}\left(\operatorname{csch}u\right)=-\operatorname{csch}u\ \coth u\cdot\frac{du}{dx}$$

Inverse Trigonometric Functions

1. $\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}},\ \forall\ |x| < 1,$
2. $\frac{d}{dx}\cos^{-1}(x) = \frac{-1}{\sqrt{1-x^2}},\ \forall\ |x| < 1,$
3. $\frac{d}{dx}\tan^{-1}(x) = \frac{1}{1+x^2},\ \forall\ x \in \mathbb{R},$
4. $\frac{d}{dx}\cot^{-1}(x) = \frac{-1}{1+x^2},\ \forall\ x \in \mathbb{R},$
5. $\frac{d}{dx}\sec^{-1}(x) = \frac{1}{|x|\sqrt{x^2-1}},\ \forall\ |x| > 1.$
6. $\frac{d}{dx}\csc^{-1}(x) = \frac{-1}{|x|\sqrt{x^2-1}},\ \forall\ |x| > 1.$

For $a > 0$,

1. $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c\ ,\ (|x| < a)$
2. $\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}dx = \sin^{-1}\left(\frac{f(x)}{a}\right) + c\ ,\ (|f| < a))$
3. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
4. $\int \frac{f'(x)}{a^2+[f(x)]^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{f(x)}{a}\right) + c$
5. $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c\ ,\ (x > a)$
6. $\int \frac{f'(x)}{f(x)\sqrt{[f(x)]^2-a^2}}dx = \frac{1}{a}\sec^{-1}\left(\frac{f(x)}{a}\right) + c\ ,\ (f > a))$

Q2: Find the derivatives of the following functions

(1) $f\ (x\ ) = \sinh^{-1}\ 2x^{4} + 1$

Solution :

$$f\ '(x\ ) = \frac{1}{\sqrt{1 + \ 2x^{4} + 1^{\ 2}}}\ 8x^{3}$$

(2) $f\ (x\ ) = \tanh^{-1}\ \ln\left|x\ \right| + 3$

Solution :

$$f\ '(x\ ) = \frac{1}{1 - \ \ln\left|x\ \right| + 3^{\ 2}}\left(\frac{1}{x}\right)$$

$$(3)\ f\ (x\ ) = 7^{\cosh^{-1} e^{2x}}$$

Solution :

$$f\ '(x\ ) = 7^{\cosh^{-1} e^{x}}\ \ln 7 \frac{1}{\sqrt{e^{x\ \ 2} - 1}} e^{x}$$

$$(4)\ f\ (x\ ) = x^{\ 5} \sinh^{-1}\ 5x$$

Solution :

$$f\ '(x\ ) = x^{\ 5} \frac{5}{\sqrt{1 + \ 5x^{\ \ 2}}} + \ 5x^{\ 4}\ \sinh^{-1}\ 5x$$

$$(5)\ f\ (x\ ) = \left[\operatorname{sech}^{-1}\ 3x\ \ + 3^{3x}\ \right]^{5}$$

Solution :

$$f\ '(x\ ) = 5\left[\operatorname{sech}^{-1}\ 3x\ \ + 3^{3x}\ \right]^{4} \frac{-1}{3x\sqrt{1-\ 3x\ ^{2}}}\ 3\ +\ 3\ \ 3^{3x}\ \ln 3$$

Q: Evaluate the following integrals

(1) $\int \frac{x}{\sqrt{x^4-16}}dx$

Solution:

$$\int \frac{x}{\sqrt{x^4-16}}dx = \frac{1}{2}\int \frac{2x}{\sqrt{(x^2)^2-4^2}}dx$$

$$= \frac{1}{2}\cosh^{-1}\left(\frac{x^2}{4}\right)+c$$

(2) $\int \frac{1}{\sqrt{4x^2+25}}dx$

Solution:

$$\int \frac{1}{\sqrt{4x^2+25}}dx = \frac{1}{2}\int \frac{2}{\sqrt{(2x)^2+5^2}}dx$$

$$= \frac{1}{2}\sinh^{-1}\left(\frac{2x}{5}\right)+c$$

(3) $\int \frac{1}{\sqrt{1+e^{2x}}}dx$

Solution :

$$\int \frac{1}{\sqrt{1+e^{2x}}}dx = \int \frac{e^{x}}{e^{x}\sqrt{1+(e^{x})^{2}}}dx$$

$$= -\operatorname{sech}^{-1} e^{x} + c$$

(4) $\int \frac{1}{x\sqrt{16-x^4}}dx$

Solution :

$$\int \frac{1}{x\sqrt{16-x^4}}dx = \int \frac{1}{x\sqrt{4^2-(x^2)^2}}dx$$

$$= \frac{1}{2}\int \frac{2x}{x^2\sqrt{4^2-(x^2)^2}}dx$$

$$= -\frac{1}{2}\times\frac{1}{4}\operatorname{sech}^{-1}\left(\frac{x^2}{4}\right)+c$$

$$= -\frac{1}{8}\operatorname{sech}^{-1}\left(\frac{x^2}{4}\right)+c$$

Q: Evaluate the following integrals

(1) $\int \frac{5x^2}{\sqrt{9+16x^6}}\,dx$

(2) $\int \frac{3x}{\sqrt{25x^4+16}}\,dx$

(3) $\int \frac{x^3}{36-9x^8}\,dx$

(4) $\int \frac{3x}{x^2\sqrt{9-16x^4}}\,dx$

(5) $\int \frac{5x}{x^2\sqrt{4-9x^4}}\,dx$

(6) $\int \frac{\sinh x}{4-25\cosh^2 x}\,dx$

(7) $\int \frac{x-2}{\sqrt{25+x^2}}\,dx$

(8) $\int \frac{x+1}{16-x^2}\,dx$

(9) $\int \frac{\cosh^{-1} x}{\sqrt{x^2-1}}\,dx$

(10) $\int \frac{2}{\sqrt{9-e^{6x}}}\,dx$

(11) $\int \frac{5}{x\sqrt{25+4x^4}}\,dx$

(12) $\int \frac{1}{x\sqrt{25-x^3}}\,dx$

(13) $\int \frac{7}{\sqrt{x^2+6x}}\,dx$

(14) $\int \frac{5}{\sqrt{x^2+2x+10}}\,dx$

(15) $\int \frac{2}{(x+3)\sqrt{x^2+6x+25}}\,dx$

(16) $\int \frac{3}{(x+1)\sqrt{3-x^2-2x}}\,dx$

The Derivatives of Hyperbolic Functions

Examples 1:

(1) $f\ (x\ ) = \cosh\left(\sqrt[3]{2x^{\ 4}+1}\right)$

Solution:

$$f\ (x\ ) = \cosh\left(\sqrt[3]{2x^{\ 4}+1}\right)$$

$$f\ '(x\ ) = \sinh\left(\sqrt[3]{2x^{\ 4}+1}\right).\left(\frac{1}{3}(2x^{\ 4}+1)^{\ -\frac{2}{3}}\right).\left(8x^{\ 3}\right)$$

(2) $f\ (x\ ) = \ln\left(\tanh\left(e^{\ -3x^{\ 4}}\right)\right)$

Solution:

$$f\ (x) = \ln\left(\tanh\left(e^{-3x^4}\right)\right)$$

$$f\ '(x) = \frac{\operatorname{sech}^2\left(e^{-3x^4}\right).\left(-12x^3 e^{-3x^4}\right)}{\tanh\left(e^{-3x^4}\right)}$$

$$= -\frac{\left(12x^3 e^{-3x^4}\right)\ \operatorname{sech}^2\left(e^{-3x^4}\right)}{\tanh\left(e^{-3x^4}\right)}$$

Integration of Hyperbolic Functions

Examples 2:

Evaluate the following integrals

$$(1)\quad \int \cosh(x)\ \operatorname{csch}^2(x)\,dx$$

Solution:

$$\int \cosh(x)\ \operatorname{csch}^2(x)\,dx$$

$$= \int \frac{\cosh(x)}{\sinh^2(x)}\,dx$$

$$Let \quad u = \sinh(x) \quad \Rightarrow du = \cosh(x)\,dx$$

$$= \int \frac{du}{u^2} \quad \Rightarrow \int u^{-2}\,du \quad \Rightarrow -u^{-1} + C$$

$$\Rightarrow -\frac{1}{u} + C = -\frac{1}{\sinh(x)} + C$$

(2) $\int \frac{\cosh(x)}{1+\sinh^2(x)}\,dx$

Solution:

$$\int \frac{\cosh(x)}{1+\sinh^2(x)}\,dx = \int \frac{\cosh(x)}{\cosh^2(x)}\,dx$$

$$\int \cosh(x)\,dx = \sinh(x)+C$$

The Derivatives of Inverse of Hyperbolic Functions

Examples 3:

(1) $f(x) = \operatorname{sech}^{-1}\left(\cos(5x^2)\right)$

Solution:

$$f(x) = \operatorname{sech}^{-1}\left(\cos(5x^2)\right)$$

$$f'(x) = -\frac{1}{\cos(5x^2)\sqrt{1-\left(\cos(5x^2)\right)^2}}.\left(-\sin(5x^2)\right).(10x)$$

(2) $f(x)=\sinh^{-1}\left(\sqrt{x}\right)$

Solution:

$$f(x)=\sinh^{-1}\left(\sqrt{x}\right)$$

$$f'(x)=\frac{1}{\sqrt{\left(\sqrt{x}\right)^2+1}}\cdot\left(\frac{1}{2}x^{-\frac{1}{2}}\right)$$

$$=\frac{1}{\sqrt{x+1}}\cdot\frac{1}{2\sqrt{x}}$$

Integration of Inverse of Hyperbolic Functions

Examples 4:

Evaluate the following integrals

(1) $\int\frac{1}{\sqrt{16+9x^2}}dx$

Solution:

$$\int \frac{1}{\sqrt{16+9x^2}}dx = \int \frac{1}{\sqrt{4^2+(3x)^2}}dx$$

$$u = 3x \Rightarrow du = 3dx \Rightarrow \frac{1}{3}du = dx$$

$$= \int \frac{1}{\sqrt{4^2+u^2}}\left(\frac{1}{3}du\right) \Rightarrow \frac{1}{3}\int \frac{1}{\sqrt{4^2+u^2}}du$$

$$\int \frac{1}{\sqrt{16+9x^2}}dx = \frac{1}{3}\sinh^{-1}\left(\frac{3x}{4}\right)+C$$

(3) $\int_4^8 \frac{x}{x^4-16}dx$

Solution:

$$\int_4^8 \frac{x}{x^4-16}dx = \int_4^8 \frac{x}{(x^2)^2-4^2}dx$$

$$u = x^2 \Rightarrow du = 2x\,dx \Rightarrow \frac{1}{2}du = x\,dx$$

$$= \int_4^8 \frac{1}{u^2-4^2}\left(\frac{1}{2}du\right) = \frac{1}{2}\int_4^8 \frac{1}{u^2-4^2}du$$

$$= \frac{1}{2}\left[-\frac{1}{4}\coth^{-1}\left(\frac{x^2}{4}\right)\right]_4^8 + C$$

$$= -\frac{1}{8}\left[\coth^{-1}\left(\frac{x^2}{4}\right)\right]_4^8 + C$$

$$= -\frac{1}{8}\left[\left[\coth^{-1}\left(\frac{8^2}{4}\right)\right]-\left[\coth^{-1}\left(\frac{4^2}{4}\right)\right]\right] + C$$

$$= -\frac{1}{8}\left[\left[\coth^{-1}\left(\frac{8^2}{4}\right)\right]-\left[\coth^{-1}\left(\frac{4^2}{4}\right)\right]\right]+C$$

$$= -\frac{1}{8}\left[\left[\coth^{-1}(16)\right]-\left[\coth^{-1}(4)\right]\right]+C$$

$$= -\frac{1}{8}\coth^{-1}(16)+\frac{1}{8}\coth^{-1}(4)+C$$

(4) $\int \frac{1}{x\sqrt{9-x^4}}dx$

Solution:

$$\int \frac{1}{x\sqrt{9-x^4}}dx = \int \frac{1}{x\sqrt{3^2-(x^2)^2}}dx$$

$$= \frac{1}{6}\operatorname{sech}^{-1}\left(\frac{x^2}{3}\right)+C$$

(5) $\int \frac{1}{\sqrt{5-e^{2x}}}dx$

Solution:

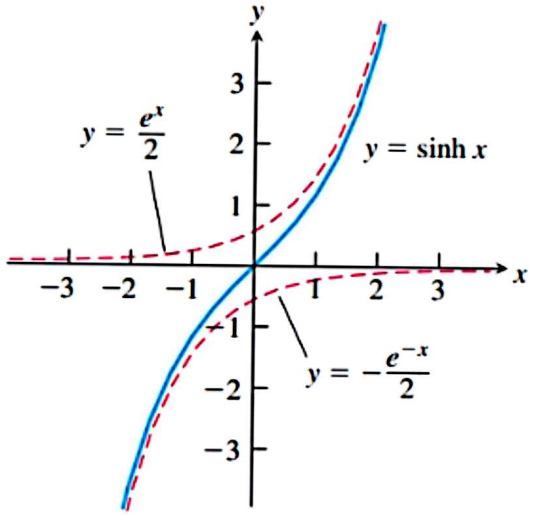
$$\int \frac{1}{\sqrt{5-e^{2x}}}dx = \int \frac{1}{\sqrt{\left(\sqrt{5}\right)^2-\left(e^x\right)^2}}dx$$

$$u = e^x \Rightarrow du = e^x\,dx$$

$$= \int \frac{e^x}{e^x\sqrt{\left(\sqrt{5}\right)^2-\left(e^x\right)^2}}dx$$

$$= \int \frac{1}{u\sqrt{\left(\sqrt{5}\right)^2-u^2}}du$$

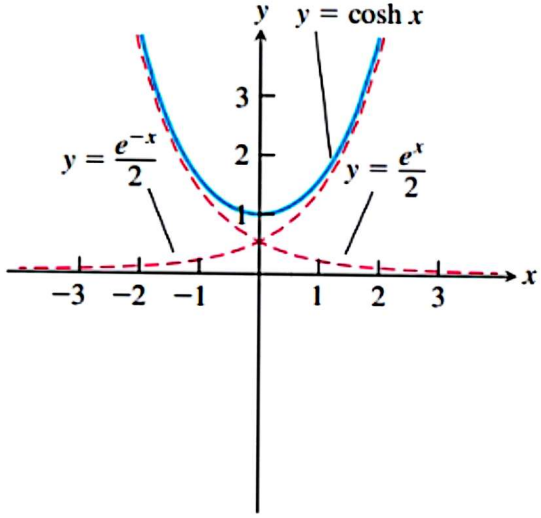
$$= -\frac{1}{\sqrt{5}}\operatorname{sech}^{-1}\left(\frac{e^x}{\sqrt{5}}\right)+C$$



(a)

Hyperbolic sine:

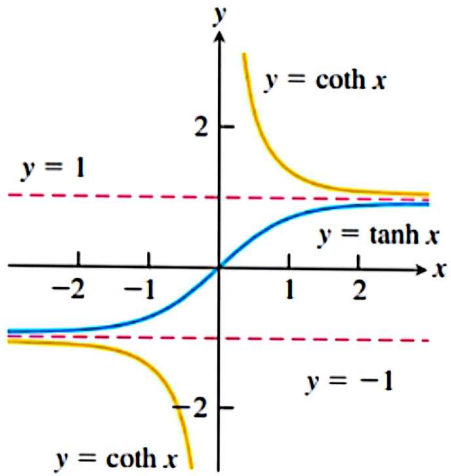
$$\sinh x = \frac{e^x - e^{-x}}{2}$$



(b)

Hyperbolic cosine:

$$\cosh x = \frac{e^x + e^{-x}}{2}$$



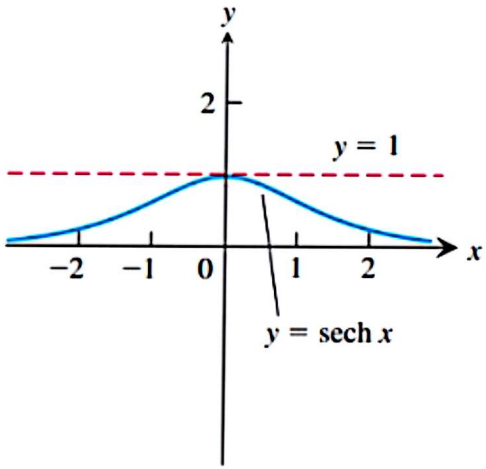
(c)

Hyperbolic tangent:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Hyperbolic cotangent:

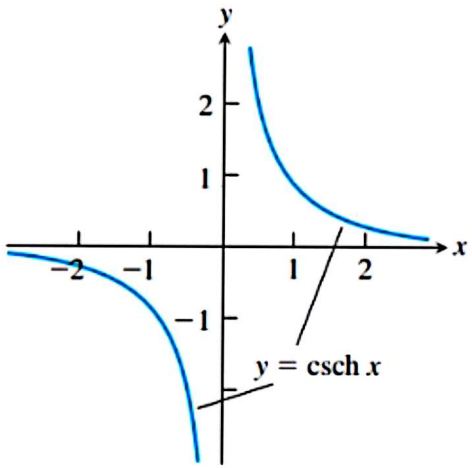
$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$



(d)

Hyperbolic secant:

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

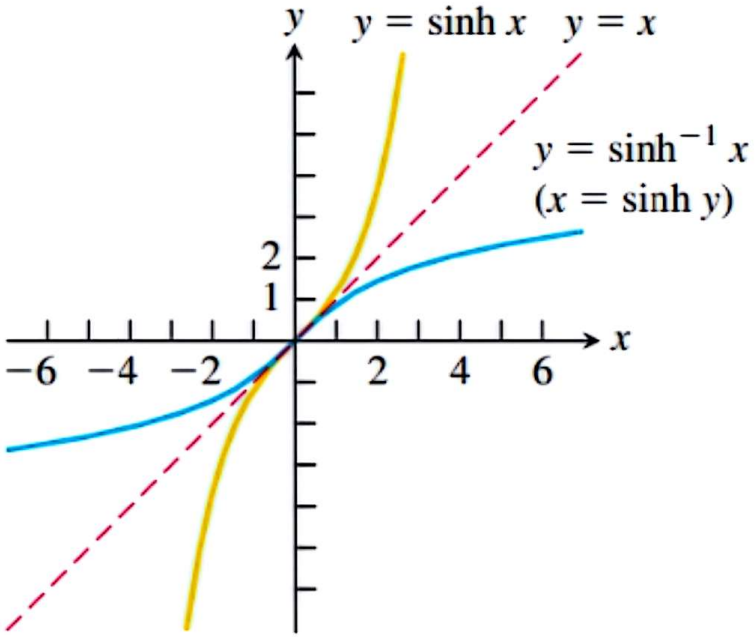


(e)

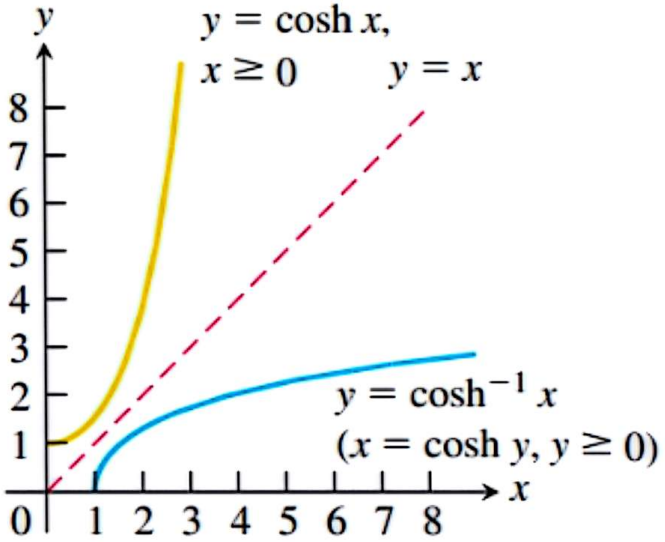
Hyperbolic cosecant:

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

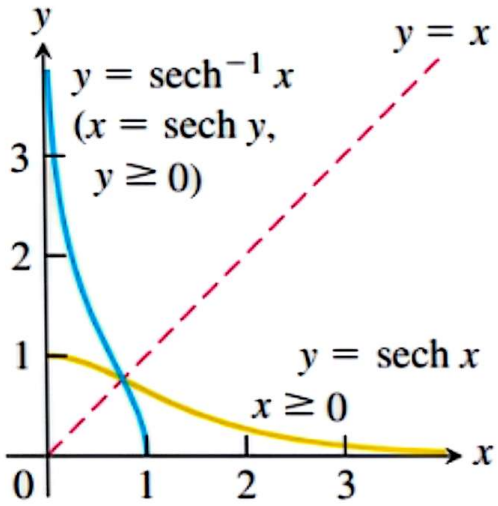
CamScanner



(a)

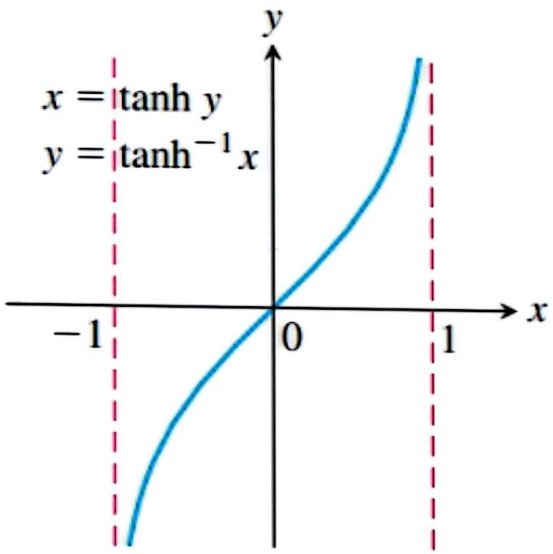


(b)

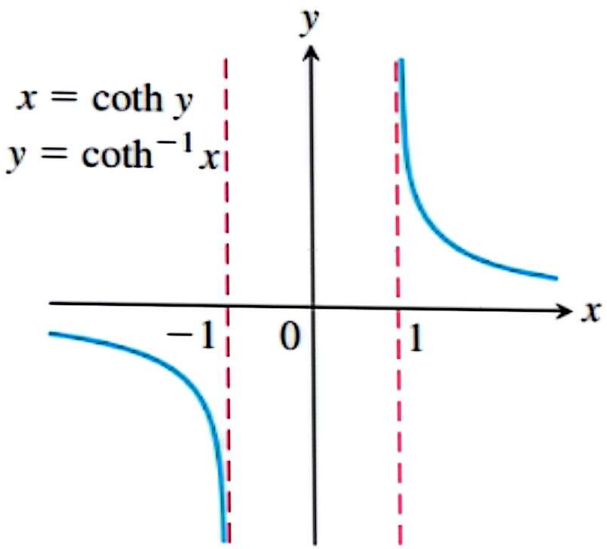


(c)

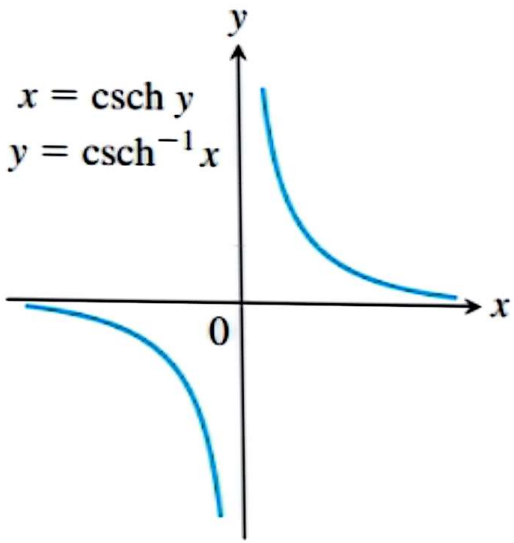
CamScanner



(a)



(b)



(c)

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Chapter 5

TECHNIQUES OF INTEGRATION

1- Integration by Parts

2- Trigonometric Integrals

3- Trigonometric Substitutions

4- Integrals of Rational Functions

5- Miscellaneous Substitutions

The methods of Integrations

(1) Integration by Part:

Theorem: If $u = f(x)$, $v = g(x)$ and if f' *and* g' are continuous, then

$$\int u\,dv = uv - \int v\,du$$

$$\mathrm{I} = uv - \int v\,du$$

Proof:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$\Rightarrow \int d(uv) = \int u\,dv + \int v\,du$$

$$Then \quad \int u\,dv = uv - \int v\,du$$

Q: Evaluate the following integrals

(1) $\int x\ \cos(x)dx$

Solution:

$$\mathrm{I}=\int x\ \cos(x)dx$$

$$\int u\,dv=uv-\int v\,du$$

$$u=x\Rightarrow du=dx$$

$$dv=\cos(x)dx\Rightarrow v=\sin(x)$$

$$\int x\ \cos(x)dx=x\ \sin(x)-\int \sin(x)dx$$

$$\int x\ \cos(x)dx=x\ \sin(x)-\ -\cos(x)\ +c$$

$$\int x\ \cos(x)dx=x\ \sin(x)+\cos(x)+c$$

(2) $\int x^2 \ln|x|dx$

Solution:

$$\int u\,dv = uv - \int v\,du$$

$$u = \ln|x| \Rightarrow du = \frac{1}{x}dx$$

$$dv = x^2 dx \Rightarrow v = \frac{1}{3}x^3$$

$$\int x^2 \ln|x|dx = \frac{1}{3}x^3 \ln|x| - \int \left(\frac{1}{3}x^3\right)\left(\frac{1}{x}dx\right)$$

$$= \frac{1}{3}x^3 \ln|x| - \frac{1}{3}\int x^3\,dx$$

$$= \frac{1}{3}x^3 \ln|x| - \frac{1}{12}x^4 + c$$

(3) $\int \tan^{-1} dx$

Solution:

$$\int u\,dv = uv - \int v\,du$$

$$u = \tan^{-1} \Rightarrow du = \frac{1}{1+x^2}dx$$

$$dv = dx \Rightarrow v = x$$

$$\int \tan^{-1} dx = x\,\tan^{-1} - \int x\left(\frac{1}{1+x^2}dx\right)$$

$$\int \tan^{-1} dx = x\,\tan^{-1} - \int \frac{x}{1+x^2}dx$$

$$\int \tan^{-1} dx = x\,\tan^{-1} - \frac{1}{2}\int \frac{2x}{1+x^2}dx$$

$$\int \tan^{-1} dx = x\,\tan^{-1} - \frac{1}{2}\ln\left|1+x^2\right| + c$$

(4) $\int e^x \cos(x)dx$

Solution :

$$\int u\,dv = uv - \int v\,du$$

$$u = \cos(x) \Rightarrow du = -\sin(x)dx$$

$$dv = e^x dx \Rightarrow v = e^x$$

$$\int e^x \cos(x)dx = e^x \cos(x) + \int e^x \sin(x)dx$$

$$\int e^x \sin(x)dx = ?$$

$$u = \sin(x) \Rightarrow du = \cos(x)dx$$

$$dv = e^x dx \Rightarrow v = e^x$$

$$\int e^x \sin(x)dx = e^x \sin(x) - \int e^x \cos(x)dx$$

$$\int e^x \cos(x)dx = e^x \cos(x) + e^x \sin(x) - \int e^x \cos(x)dx$$

$$\int e^x \cos(x)dx + \int e^x \cos(x)dx = e^x \cos(x) + e^x \sin(x)$$

$$2\int e^x \cos(x)dx = e^x \cos(x) + e^x \sin(x)$$

$$\int e^x \cos(x)dx = \frac{1}{2}e^x \cos(x) + \frac{1}{2}e^x \sin(x)$$

(5) $\int x^2 e^x dx$

Solution:

$$\int u\,dv = uv - \int v\,du$$

$$u = x^2 \Rightarrow du = 2x\,dx$$

$$dv = e^x dx \Rightarrow v = e^x$$

$$\int x^2 e^x dx = x^2 e^x - \int e^x \; 2x\,dx$$

$$\int x^2 e^x dx = x^2 e^x - 2\int x e^x dx$$

$$u = x \Rightarrow du = dx$$

$$dv = e^x dx \Rightarrow v = e^x$$

$$\int x^2 e^x dx = x^2 e^x - 2\left[x e^x - \int e^x dx\right]$$

$$\int x^2 e^x dx = x^2 e^x - 2\left[x e^x - e^x\right]$$

$$\int x^2 e^x dx = x^2 e^x - 2x e^x - 2e^x$$

$$\int x^2 e^x dx = x^2 e^x - 2x e^x - 2e^x + c$$

(16) $\int \ln(x)^2 dx$

Solution :

$$\int u\,dv = uv - \int v\,du$$

$$u = \ln(x)^2 \Rightarrow du = 2\frac{1}{x}\ln(x)dx$$

$$dv = dx \Rightarrow v = x$$

$$\int \ln(x)^2 dx = x\ln(x)^2 - \int x\left(2\frac{1}{x}\ln(x)dx\right)$$

$$\int \ln(x)^2 dx = x\ln(x)^2 - 2\int \ln(x)dx$$

$$\int \ln(x)dx = ?$$

$$u = \ln(x) \Rightarrow du = \frac{1}{x}dx$$

$$dv = dx \Rightarrow v = x$$

$$\int \ln(x)dx = x\ln(x) - \int x\left(\frac{1}{x}dx\right)$$

$$\int \ln(x)dx = x\ln(x) - \int dx$$

$$\int \ln(x)dx = x\ln(x) - x$$

$$\int \left(\ln(x)\right)^2 dx = x\left(\ln(x)\right)^2 - 2\int \ln(x)dx$$

$$\int \left(\ln(x)\right)^2 dx = x\left(\ln(x)\right)^2 - 2\left[x\ln(x) - x\right]$$

$$\int \left(\ln(x)\right)^2 dx = x\left(\ln(x)\right)^2 - 2x\ln(x) + 2x + c$$

Q: Evaluate the following integrals

(1) $\int (2x+1)\cosh x\,dx$

(2) $\int (x^2-1)\sin 2x\,dx$

(3) $\int x^3 e^x\,dx$

(4) $\int (x+1)\sec^2 x\,dx$

(5) $\int \ln|x|\,dx$

(6) $\int x^5 \ln|x|\,dx$

(7) $\int x^{-4}\ln|x|\,dx$

(8) $\int \sec^{-1} x\,dx$

(9) $\int \sinh^{-1} x\,dx$

(10) $\int x\,csch^{-1}x\,dx$

(11) $\int e^{2x}\sin 3x\,dx$

(12) $\int e^{5x}\cos 2x\,dx$

(13) $\int e^{3x}\sinh x\,dx$

(14) $\int e^{x}\cosh 5x\,dx$

(15) $\int e^{\sqrt{x}}\,dx$

(16) $\int (\ln|x|)^2\,dx$

(17) $\int x^3\sqrt{x^3-1}\,dx$

(18) $\int x^7\sqrt{x^4+2}\,dx$

Trigonometric Integrals

INTEGRALS INVOLVING TRIGONOMETRIC FUNCTIONS

(1) $\int \sin^n(x)\,dx$

(2) $\int \cos^m(x)\,dx$

(3) $\int \sin^n(x)\cos^m(x)\,dx$

(4) $\int \sin(ax)\cos(bx)\,dx$

(5) $\int \sin^n(x)\sin^m(x)\,dx$

(6) $\int \cos^n(x)\cos^m(x)\,dx$

(7) $\int \sec^m(x)\,dx$

(8) $\int \tan^n(x)\,dx$

(9) $\int \tan^n(x)\sec^m(x)\,dx$

1. $\int \sin(ax)\sin(bx)dx = \frac{1}{2}\int \cos((a-b)x) - \cos((a+b)x)dx,$
2. $\int \sin(ax)\cos(bx)dx = \frac{1}{2}\int \sin((a+b)x) + \sin((a-b)x)dx,$
3. $\int \cos(ax)\cos(bx)dx = \frac{1}{2}\int \cos((a+b)x) + \cos((a-b)x)dx,$

For $a, b \in \mathbb{R}$ such that $|a| \neq |b|$, we have

4. $2\int \sin(ax)\sin(bx)dx = \frac{\sin((a-b)x)}{a-b} - \frac{\sin((a+b)x)}{a+b} + c,$
5. $2\int \cos(ax)\cos(bx)dx = \frac{\sin((a+b)x)}{a+b} + \frac{\sin((a-b)x)}{a-b} + c,$
6. $2\int \sin(ax)\cos(bx)dx = -\frac{\cos((a+b)x)}{a+b} - \frac{\cos((a-b)x)}{a-b} + c,$

$$\int \sinh(ax)\sinh(bx)dx = \frac{1}{2}\int \cosh((a+b)x) - \cosh((a-b)x)dx,$$

$$\int \sinh(ax)\cosh(bx)dx = \frac{1}{2}\int \sinh((a+b)x) + \sinh((a-b)x)dx,$$

$$\int \cosh(ax)\cosh(bx)dx = \frac{1}{2}\int \cosh((a+b)x) + \cosh((a-b)x)dx,$$

1. $\cos^2(x) + \sin^2(x) = 1,$
2. $\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y),$
3. $\sin(x-y) = \sin(x)\cos(y) - \cos(x)\sin(y),$
4. $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y),$
5. $\cos(x-y) = \cos(x)\cos(y) + \sin(x)\sin(y),$
6. $\sin(x)\cos(y) = \frac{1}{2}(\sin(x+y) + \sin(x-y)),$
7. $\sin(x)\sin(y) = \frac{1}{2}(\cos(x-y) - \cos(x+y)),$
8. $\cos(x)\cos(y) = \frac{1}{2}(\cos(x+y) + \cos(x-y)),$
9. $\cos(x) + \cos(y) = 2\cos(\frac{x+y}{2})\cos(\frac{x-y}{2}),$
10. $\cos(x) - \cos(y) = -2\sin(\frac{x+y}{2})\sin(\frac{x-y}{2}),$
11. $\sin(x) + \sin(y) = 2\sin(\frac{x+y}{2})\cos(\frac{x-y}{2}),$

12. $\sin(x) - \sin(y) = 2\cos(\frac{x+y}{2})\sin(\frac{x-y}{2})$,
13. $\cos(2x) = \cos^2(x) - \sin^2(x)$,
14. $\cos^2(x) = \dfrac{1+\cos(2x)}{2}$,
15. $\sin^2(x) = \dfrac{1-\cos(2x)}{2}$,
16. $\sec^2(x) = 1 + \tan^2(x)$,
17. $\csc^2(x) = 1 + \cot^2(x)$,
18. $\tan(x+y) = \dfrac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}$,
19. $\tan(x-y) = \dfrac{\tan(x)-\tan(y)}{1+\tan(x)\tan(y)}$,

Some Important Hyperbolic Formulas

1. $\cosh^2(x) - \sinh^2(x) = 1$,
2. $\sinh(x+y) = \sinh(x)\cosh(y) + \cosh(x)\sinh(y)$,
3. $\sinh(x-y) = \sinh(x)\cosh(y) - \cosh(x)\sinh(y)$,
4. $\cosh(x+y) = \cosh(x)\cosh(y) + \sinh(x)\sinh(y)$,
5. $\cosh(x-y) = \cosh(x)\cosh(y) - \sinh(x)\sinh(y)$,
6. $\sinh(x)\cosh(y) = \dfrac{1}{2}(\sinh(x+y) + \sinh(x-y))$,
7. $\sinh(x)\sinh(y) = \dfrac{1}{2}(\cosh(x+y) + \cosh(x-y))$,
8. $\cosh(x)\cosh(y) = \dfrac{1}{2}(\cosh(x+y) - \cosh(x-y))$,

Trigonometric Integrals

1- Evaluate the Following Integrals

(1) $\int \sin^2 x\, dx$ (2) $\int \sin^3 x\, dx$

(3) $\int \cos^3 x\, dx$ (4) $\int \cos^2 x\, dx$

(5) $\int \sin^5 x\, dx$ (6) $\int \sin^4 x\, dx$

(7) $\int \cos^4 x\, dx$

(1) $\int \sin^2 x\, dx$

Solution :

$$\int \sin^2 x\, dx = \frac{1}{2}\int (1-\cos 2x)\, dx$$

$$= \frac{1}{2}\left[x - \frac{1}{2}\sin 2x\right] + c$$

$$= \frac{1}{2}x - \frac{1}{4}\sin 2x + c$$

(3) $\int \cos^3 x \, dx$

Solution :

$$\int \cos^3 x \, dx = \int \cos^2 x \, \cos^1 x \, dx$$

$$= \int (1 - \sin^2 x) \cos x \, dx$$

$$u = \sin x \quad \Rightarrow \quad du = \cos x \, dx$$

$$= \int (1 - u^2) \, du$$

$$= u - \frac{1}{3} u^3 + c$$

$$= \sin x - \frac{1}{3} \sin^3 x + c$$

(5) $\int \sin^5 x\, dx$

Solution :

$$\int \sin^5 x\, dx = \int \sin^4 x \sin x\, dx$$

$$= \int (\sin^2 x)^2 \sin x\, dx$$

$$= \int (1-\cos^2 x)^2 \sin x\, dx$$

$$u = \cos x \quad \Rightarrow \quad du = -\sin x\, dx$$

$$= -\int (1-u^2)^2\, du$$

$$= -\int (1-2u^2+u^4)\, du$$

$$= -\left|u - \frac{2}{3}u^3 + \frac{1}{5}u^5\right| + c$$

$$= -u + \frac{2}{3}u^3 - \frac{1}{5}u^5 + c$$

$$\int \sin^5 x\, dx = -\cos x + \frac{2}{3}(\cos x)^3 - \frac{1}{5}(\cos x)^5 + c$$

(6) $\int \sin^4 x\, dx$

Solution :

$$\int \sin^4 x\, dx = \int \sin^2 x \cdot \sin^2 x\, dx$$

$$= \int (\sin^2 x)^2\, dx = \int \left[\frac{1}{2}(1-\cos 2x)\right]^2 dx$$

$$= \frac{1}{4}\int (1-\cos 2x)^2\, dx$$

$$= \frac{1}{4}\int (1-2\cos 2x + \cos^2 2x)\, dx$$

$$\cos^2 2x = \frac{1}{2}(1+\cos 4x) = \frac{1}{2}+\frac{1}{2}\cos 4x$$

$$= \frac{1}{4}\int \left(1-2\cos 2x + \frac{1}{2}+\frac{1}{2}\cos 4x\right) dx$$

$$= \frac{1}{4}\int \left(\frac{3}{2}-2\cos 2x + \frac{1}{2}\cos 4x\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x - \sin 2x + \frac{1}{8}\sin 4x\right] + c$$

$$\int \sin^4 x\, dx = \frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c$$

2- Evaluate the Following Integrals

(1) *Odd and Odd*

(1) $\int \sin^5 x \cos^3 x \, dx$

(2) $\int \cos^5 x \sin^3 x \, dx$

(3) $\int \sin^7 x \cos^5 x \, dx$

(1) $\int \sin^5 x \cos^3 x \, dx$

Solution :

$$\int \sin^5 x \cos^3 x \, dx = \int \sin^5 x \cdot \cos^2 x \cdot \cos x \, dx$$

$$= \int \sin^5 x \cdot (1 - \sin^2 x) \cdot \cos x \, dx$$

$$u = \sin x \Rightarrow du = \cos x \, dx$$

$$= \int u^5 \cdot (1 - u^2) du$$

$$= \int (u^5 - u^7) du$$

$$= \frac{1}{6} u^6 - \frac{1}{8} u^8 + c$$

$$\int \sin^5 x \cos^3 x \, dx = \frac{1}{6} \sin^6 x - \frac{1}{8} \sin^8 x + c$$

(2) *Odd and Even*

(1) $\int \sin^5 x \cos^4 x \, dx$

(2) $\int \cos^5 x \sin^4 x \, dx$

(3) $\int \cos^7 x \sin^6 x \, dx$

(1) $\int \sin^5 x \cos^4 x \, dx$

Solution :

$$\int \sin^5 x \cos^4 x \, dx = \int \sin^4 x \cos^4 x \sin x \, dx$$

$$= \int (\sin^2 x)^2 \cos^4 x \sin x \, dx$$

$$= \int (1-\cos^2 x)^2 \cos^4 x \sin x \, dx$$

$$u = \cos x \quad \Rightarrow \quad du = -\sin x \, dx$$

$$= -\int (1-u^2)^2 u^4 \, du$$

$$= -\int (1-2u^2+u^4)\cdot u^4 \, du$$

$$= -\int (u^4 - 2u^6 + u^8) \, du$$

$$= -\left[\frac{1}{5}u^5 - \frac{2}{7}u^7 + \frac{1}{9}u^9\right] + c$$

$$= -\frac{1}{5}u^5 + \frac{2}{7}u^7 - \frac{1}{9}u^9 + c$$

$$= -\frac{1}{5}\cos^5 x + \frac{2}{7}\cos^7 x - \frac{1}{9}\cos^9 x + c$$

(3) *Even and Even*

(1) $\int \sin^2 x \cos^2 x \, dx$ (2) $\int \sin^4 x \cos^2 x \, dx$

(3) $\int \cos^6 x \sin^2 x \, dx$ (4) $\int \sin^6 x \cos^2 x \, dx$

(5) $\int \cos^4 x \sin^2 x \, dx$

(1) $\int \sin^2 x \cos^2 x \, dx$

Solution :

$$= \int \frac{1}{2}(1-\cos 2x)\cdot\frac{1}{2}(1+\cos 2x)\,dx$$

$$= \int \frac{1}{4}(1-\cos 2x)\cdot(1+\cos 2x)\,dx$$

$$= \int \frac{1}{4}(1-\cos^2 2x)\,dx$$

$$= \int \left[\frac{1}{4}-\frac{1}{4}\cos^2 2x\right]dx$$

$$= \int \left[\frac{1}{4}-\frac{1}{4}\left(\frac{1}{2}(1+\cos 4x)\right)\right]dx$$

$$= \int \left[\frac{1}{4}-\frac{1}{8}-\frac{1}{8}\cos 4x\right]dx$$

$$= \int \left[\frac{1}{8} - \frac{1}{8} \cos 4x \right] dx$$

$$= \frac{1}{8} x - \frac{1}{8} \cdot \frac{1}{4} \sin 4x + c$$

$$\int \sin^2 x \cos^2 x \, dx = \frac{1}{8} x - \frac{1}{32} \sin 4x + c$$

Trigonometric Integral of $\sec x$

Evaluate the following integrals

(1) $\int \sec x \, dx$

(2) $\int \sec^2 x \, dx$

(3) $\int \sec^3 x \, dx$

(4) $\int \sec^4 x \, dx$

(5) $\int \tan x \, dx$

(6) $\int \tan^2 x \, dx$

(7) $\int \tan^3 x \, dx$

(8) $\int \tan^4 x \, dx$

(4) $\int \sec^4 x\, dx$

Solution :

$$\int \sec^4 x\, dx = \int \sec^2 x \cdot \sec^2 x\, dx$$

$$= \int (1+\tan^2 x)\cdot \sec^2 x\, dx$$

$$u = \tan x \quad \Rightarrow \quad du = \sec^2 x\, dx$$

$$= \int (1+u^2)du$$

$$= u + \frac{1}{3}u^3 + c$$

$$\int \sec^4 x\, dx = \tan x + \frac{1}{3}\tan^3 x + c$$

(3) $\int \tan^3 x \, dx$

Solution :

$$\int \tan^3 x \, dx = \int \tan^2 x \ \tan x \, dx$$

$$= \int (\sec^2 x - 1) \tan x \, dx$$

$$= \int (\sec^2 x \cdot \tan x - \tan x) \, dx$$

$$= \int \tan x \sec^2 x \, dx - \int \tan x \ dx$$

$$u = \tan x \ \Rightarrow \ du = \sec^2 x \, dx$$

$$= \int \tan x \sec^2 x \, dx - \int \tan x \ dx$$

$$= \int u \, du + \ln|\cos x| + c$$

$$= \frac{1}{2} u^2 + \ln|\cos x| + c$$

$$\int \tan^3 x \, dx = \frac{1}{2} \tan^2 x + \ln|\cos x| + c$$

Evaluate the following integral

(1) $\int \tan^3 x \cdot \sec^4 x \, dx$

(2) $\int \tan^4 x \cdot \sec^3 x \, dx$

(3) $\int \tan^5 x \cdot \sec x \, dx$

(1) $\int \tan^3 x \cdot \sec^4 x \, dx$

Solution :

$$\int \tan^3 x \cdot \sec^4 x \, dx = \int \tan^3 x \cdot \sec^2 x \cdot \sec^2 x \, dx$$

$$= \int \tan^3 x \cdot (1 + \tan^2 x) \cdot \sec^2 x \, dx$$

$$u = \tan x \quad \Rightarrow \quad du = \sec^2 x \, dx$$

$$= \int u^3 (1 + u^2) du = \int (u^3 + u^5) du$$

$$= \frac{1}{4} u^4 + \frac{1}{6} u^6 + c$$

$$\int \tan^3 x \cdot \sec^4 x \, dx = \frac{1}{4} \tan^4 x + \frac{1}{6} \tan^6 x + c$$

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Integrals of Product Trigonometric Functions

$$(1)\ \sin(u)\cos(v) = \frac{1}{2}[\sin(u+v)+\sin(u-v)]$$

$$(2)\ \cos(u)\sin(v) = \frac{1}{2}[\sin(u+v)-\sin(u-v)]$$

$$(3)\ \cos(u)\cos(v) = \frac{1}{2}[\cos(u+v)+\cos(u-v)]$$

$$(4)\ \sin(u)\sin(v) = \frac{1}{2}[\cos(u-v)-\cos(u+v)]$$

Evaluate the following integrals

$$(1)\ \int \sin(5x)\sin(3x)\,dx$$

$$(2)\ \int \sin(4x)\cos(3x)\,dx$$

$$(3)\ \int \cos(4x)\cos(8x)\,dx$$

$$(4)\ \int \cos(5x)\sin(3x)\,dx$$

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(1) $\int \sin(5x)\sin(3x)dx$

Solution :

$$\int \sin(5x)\sin(3x)dx$$

$$= \frac{1}{2}\int [\cos(5x - 3x) - \cos(5x + 3x)]$$

$$= \frac{1}{2}\int [\cos(2x) - \cos(8x)]$$

$$= \frac{1}{2}\left[\frac{1}{2}\sin(2x) - \frac{1}{8}\sin(8x)\right] + c$$

$$= \frac{1}{4}\sin(2x) - \frac{1}{16}\sin(8x) + c$$

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(2) $\int \sin(4x)\cos(3x)dx$

Solution :

$$\int \sin(4x)\cos(3x)dx$$

$$= \frac{1}{2}\int [\sin(4x + 3x) + \sin(4x - 3x)]$$

$$= \frac{1}{2}\int [\sin(7x) + \sin(x)]$$

$$= \frac{1}{2}\left[-\frac{1}{7}\cos(7x) - \cos(x)\right] + c$$

$$= -\frac{1}{14}\cos(7x) - \frac{1}{2}\cos(x) + c$$

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(6) $\int \tan^5(x)\sec^5(x)\,dx$

Solution :

$$\int \tan^4(x)\sec^4(x)\sec(x)\tan(x)\,dx$$

$$= \int \left(\tan^2(x)\right)^2 \left(\sec^2(x)\right)^2 \sec(x)\tan(x)\,dx$$

$$= \int \left(\sec^2(x)-1\right)^2 \left(\sec^2(x)\right)^2 \sec(x)\tan(x)\,dx$$

$$u = \sec(x) \Rightarrow du = \sec(x)\tan(x)\,dx$$

$$\int \left(\sec^2(x)-1\right)^2 \left(\sec^2(x)\right)^2 \sec(x)\tan(x)\,dx$$

$$= \int \left(u^2-1\right)^2 \left(u^2\right)^2 du$$

$$= \int \left(u^4-2u^2+1\right)u^4\,du$$

$$= \int \left(u^8-2u^6+u^4\right)du$$

$$= \frac{1}{9}u^9 - \frac{2}{7}u^7 + \frac{1}{5}u^5 + c$$

$$= \frac{1}{9}\sec^9(x) - \frac{2}{7}\sec^7(x) + \frac{1}{5}\sec^5(x) + c$$

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(9) $\int \cot^5(x)\csc^3(x)dx$

Solution :

$$\int \cot^4(x)\csc^2(x)\csc(x)\cot(x)dx$$

$$=\int \left(\cot^2(x)\right)^2 \csc^2(x)\csc(x)\cot(x)dx$$

$$=\int \left(\csc^2(x)-1\right)^2 \csc^2(x)\csc(x)\cot(x)dx$$

$$u=\csc(x) \Rightarrow du=-\csc(x)\cot(x)dx$$

$$\int \left(\csc^2(x)-1\right)^2 \csc^2(x)\csc(x)\cot(x)dx$$

$$=-\int \left(u^2-1\right)^2 u^2 du$$

$$=-\int \left(u^4-2u^2+1\right)u^2 du$$

$$=-\int \left(u^6-2u^4+u^2\right)du$$

$$=-\left[\frac{1}{7}u^7-\frac{2}{5}u^5+\frac{1}{3}u^3\right]+c$$

$$=-\frac{1}{7}u^7+\frac{2}{5}u^5-\frac{1}{3}u^3+c$$

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د. موسى رحمه جادالله عجال
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Evaluate the following integrals:

(1) $\int \sin^3 x \cos^4 x \, dx$

(2) $\int \sin^2 x \cos^5 x \, dx$

(3) $\int \sin^5 x \cos^3 x \, dx$

(4) $\int \sqrt[3]{\cos x} \sin^3 x \, dx$

(5) $\int \sqrt{\tan x} \sec^4 x \, dx$

(6) $\int \tan^5 x \sec^5 x \, dx$

(7) $\int \tan^3 x \sec^4 x \, dx$

(8) $\int \cot^2 x \csc^6 x \, dx$

(9) $\int \cot^5 x \csc^3 x \, dx$

(10) $\int \sin 3x \cos 5x \, dx$

(11) $\int \sin 2x \sin 6x \, dx$

(12) $\int \cos 4x \cos 7x \, dx$

(13) $\int \sin^2 x \, dx$

(14) $\int \cos^2 x \, dx$

(15) $\int \sec^5 x \, dx$

(16) $\int \tan^6 x \, dx$

Wondershare PDFelement

3- Trigonometric Substitutions

Trigonometric Substitutions

(1) $\sqrt{\alpha^2 - x^2} \Rightarrow x = \alpha \sin(\theta)$

$\Rightarrow x = \alpha \cos(\theta)$

(2) $\sqrt{\alpha^2 + x^2} \Rightarrow x = \alpha \sec(\theta)$

(3) $\sqrt{x^2 - \alpha^2} \Rightarrow x = \alpha \tan(\theta)$

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(1) $\int \frac{1}{x\sqrt{4-x^2}}dx$

Solution :

Let $x = a\sin\theta$

$a = 2, \quad x = 2\sin\theta, \Rightarrow dx = 2\cos\theta d\theta$

$\sqrt{4-x^2} = \sqrt{4-(2\sin\theta)^2} = \sqrt{4-4\sin^2\theta}$

$= \sqrt{4(1-\sin^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta$

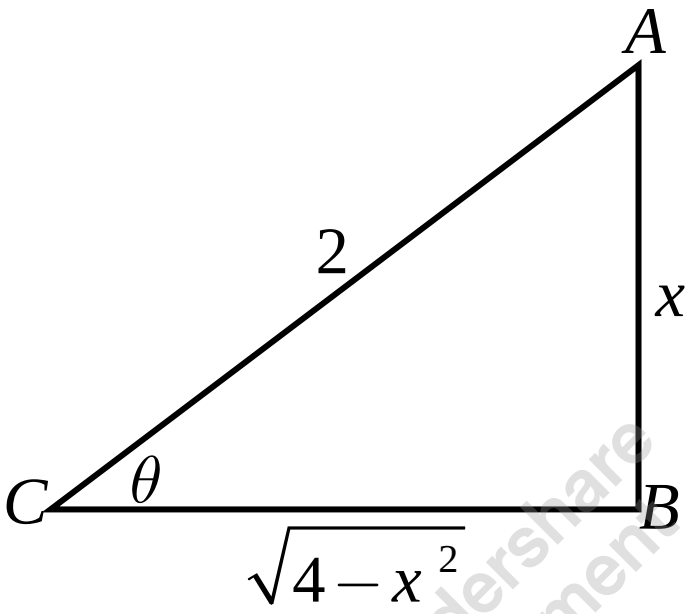
$\int \frac{1}{x\sqrt{4-x^2}}dx = \int \frac{2\cos\theta d\theta}{2\sin\theta \cdot 2\cos\theta}$

$= \int \frac{d\theta}{2\sin\theta} = \frac{1}{2}\int \frac{1}{\sin\theta}d\theta = \frac{1}{2}\int \csc\theta d\theta$

$= \frac{1}{2}\ln|\csc\theta - \cot\theta| + c$

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$$x = 2\sin\theta \quad \Rightarrow \quad \sin\theta = \frac{x}{2}$$



$$\csc\theta = \frac{2}{x}, \quad \cot\theta = \frac{\sqrt{x - x^2}}{x}$$

$$\int \frac{1}{x\sqrt{4-x^2}}dx = \frac{1}{2}\ln\left|\frac{2}{x} - \frac{\sqrt{x-x^2}}{x}\right| + c$$

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(2) $\int \frac{1}{x^2\sqrt{x^2+9}}dx$

Solution :

Let $x = a\tan\theta$

$a=3, \quad x = 3\tan\theta \quad \Rightarrow dx = 3\sec^2\theta d\theta$

$$\int \frac{1}{x^2\sqrt{x^2+9}}dx = \int \frac{3\sec^2\theta d\theta}{(3\tan\theta)^2\sqrt{(3\tan\theta)^2+9}}$$

$$= \int \frac{3\sec^2\theta d\theta}{9\tan^2\theta\sqrt{9\tan^2\theta+9}}$$

$$= \int \frac{3\sec^2\theta d\theta}{9\tan^2\theta\sqrt{9(\tan^2\theta+1)}}$$

$$= \int \frac{3\sec^2\theta d\theta}{9\tan^2\theta\sqrt{9\sec^2\theta}} = \int \frac{3\sec^2\theta d\theta}{9\tan^2\theta\cdot 3\sec\theta}$$

$$= \int \frac{\sec\theta d\theta}{9\tan^2\theta} = \frac{1}{9}\int \frac{\sec\theta}{\tan^2\theta}d\theta = \frac{1}{9}\int \frac{\cos\theta}{\sin^2\theta}d\theta$$

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$$\frac{1}{9}\int \frac{\cos\theta}{\sin^2\theta}d\theta = ?$$

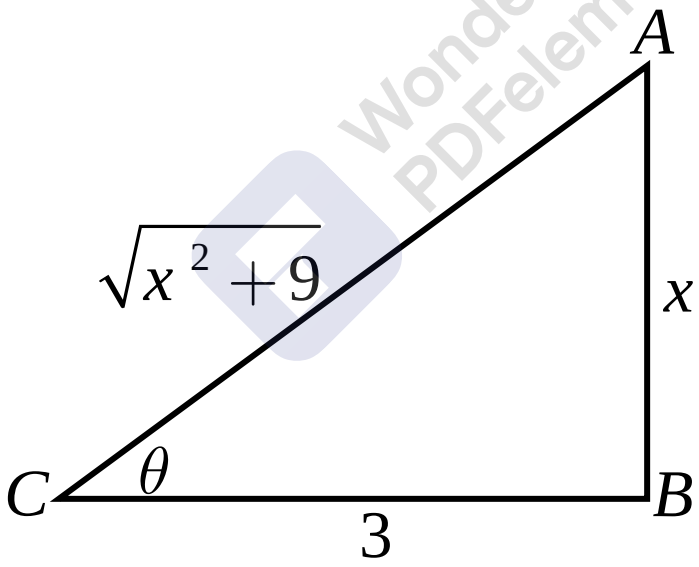
$$u = \sin\theta \;\Rightarrow\; du = \cos\theta\, d\theta$$

$$\frac{1}{9}\int \frac{\cos\theta}{\sin^2\theta}d\theta = \frac{1}{9}\int \frac{1}{u^2}du = \frac{1}{9}\int u^{-2}\,du$$

$$= -\frac{1}{9}u^{-1} + c = -\frac{1}{9}(\sin\theta)^{-1} + c$$

$$= -\frac{1}{9}\frac{1}{\sin\theta} + c = -\frac{1}{9}\csc\theta + c$$

$$x = 3\tan\theta \quad \Rightarrow \quad \tan\theta = \frac{x}{3}$$



$$\csc\theta = \frac{\sqrt{x^2+9}}{x}$$

$$\int \frac{1}{x^2\sqrt{x^2+9}}dx = -\frac{1}{9}\frac{\sqrt{x^2+9}}{x} + c$$

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(3) $\int \frac{1}{x^2\sqrt{x^2-25}}dx$

Solution :

Let $x = a\sec\theta$

$a=5, \quad x=5\sec\theta \Rightarrow dx = 5\sec\theta\tan\theta\, d\theta$

$$\int \frac{1}{x^2\sqrt{x^2-25}}dx =$$

$$= \int \frac{5\sec\theta\tan\theta\, d\theta}{(5\sec\theta)^2\sqrt{(5\sec\theta)^2-25}}$$

$$= \int \frac{5\sec\theta\tan\theta\, d\theta}{25\sec^2\theta\sqrt{25\sec^2\theta-25}}$$

$$= \int \frac{5\sec\theta\tan\theta\, d\theta}{25\sec^2\theta\sqrt{25(\sec^2\theta-1)}}$$

$$= \int \frac{5\sec\theta\tan\theta\, d\theta}{25\sec^2\theta\sqrt{25\tan^2\theta}}$$

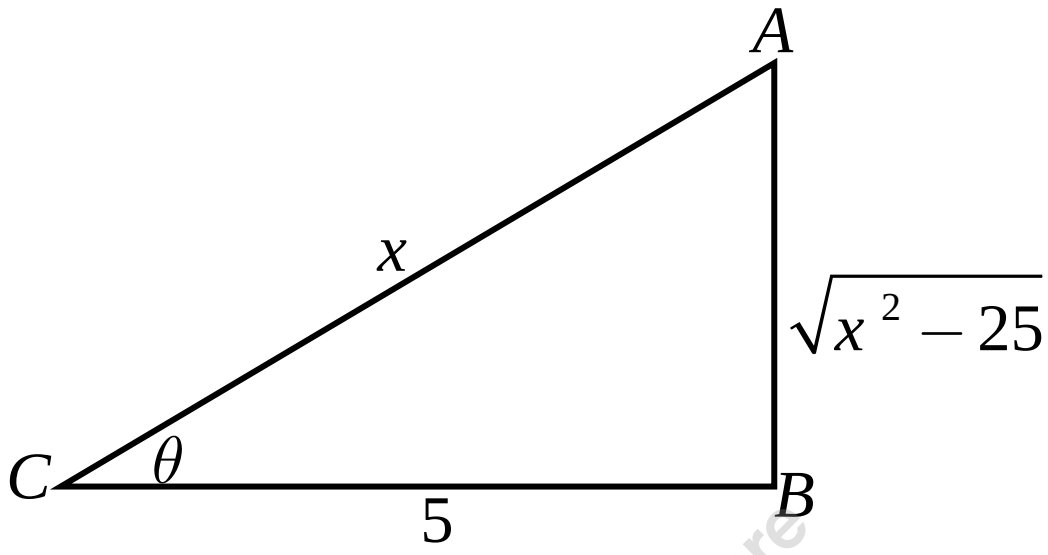
$$= \int \frac{5\sec\theta\tan\theta\, d\theta}{25\sec^2\theta\cdot 5\tan\theta}$$

$$= \int \frac{d\theta}{25\sec\theta} = \frac{1}{25}\int \frac{1}{\sec\theta}\, d\theta = \frac{1}{25}\int \cos\theta\, d\theta$$

$$= \frac{1}{25}\sin\theta + c$$

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$$x = 5\sec\theta \quad \Rightarrow \quad x = \sec\theta = \frac{x}{5}$$



$$\sin\theta = \frac{\sqrt{x^2-25}}{x}$$

$$\int \frac{1}{x^2\sqrt{x^2-25}}dx = \frac{1}{25}\sin\theta + c$$

$$= \frac{1}{25}\frac{\sqrt{x^2-25}}{x} + c$$

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Home work:

Evaluate the following integrals

(1) $\int \frac{1}{x\sqrt{25x^2+16}}dx$

(2) $\int \frac{x}{\sqrt{x^2+9}}dx$

(3) $\int \frac{x^2}{\sqrt{x^2-16}}dx$

(4) $\int x^3\sqrt{16-x^2}\,dx$

(5) $\int (1-2x^2)^{\frac{3}{2}}dx$

(6) $\int \frac{x}{\sqrt{2x-x^2}}dx$

(7) $\int \frac{(x-2)^3}{\sqrt{4x-x^2}}dx$

(8) $\int \sqrt{10x-x^2}\,dx$

(9) $\int \frac{x^3}{(3+x^2)^{\frac{5}{2}}}dx$

Wondershare PDFelement

4- Example:

Evaluate $\int \frac{\sin 2x}{\sin^2 x - 2\sin x - 8} dx$

$$\int \frac{\sin 2x}{\sin^2 x - 2\sin x - 8} dx$$

Solution :

$$\int \frac{\sin 2x}{\sin^2 x - 2\sin x - 8} dx = \int \frac{2\sin x \cos x}{\sin^2 x - 2\sin x - 8} dx$$

$$u = \sin x \quad \Rightarrow \quad du = \cos x \, dx$$

$$= \int \frac{2u}{u^2 - 2u - 8} du = \int \frac{2u}{(u-4)(u+2)} du$$

$$= \frac{A}{(u-4)} + \frac{B}{(u+2)}$$

$$\frac{2u}{(u-4)(u+2)} = \frac{A(u+2) + B(u-4)}{(u-4)(u+2)}$$

$$2u = A(u+2) + B(u-4)$$

$$Let \ u = -2 \quad \Rightarrow \quad 2(-2) = A(-2+2) + B(-2-4)$$

$$-4 = -6B \quad \Rightarrow \quad -4 = -6B \ \Rightarrow \ B = \frac{-4}{-6} \ \Rightarrow \ B = \frac{2}{3}$$

$$2u = A(u+2) + B(u-4)$$

$$Let\ u = 4 \quad \Rightarrow \quad 2(4) = A(4+2) + B(4-4)$$

$$8 = 6A \ \Rightarrow A = \frac{8}{6} \ \Rightarrow \ A = \frac{4}{3}$$

$$= \int \frac{2u}{u^2 - 2u - 8} du = \frac{A}{(u-4)} + \frac{B}{(u+2)}$$

$$= \int \left[\frac{\frac{4}{3}}{(u-4)} + \frac{\frac{2}{3}}{(u+2)} \right] du = \frac{4}{3}\ln(u-4) + \frac{2}{3}\ln(u+2) + c$$

$$= \frac{4}{3}\ln(\sin(x)-4) + \frac{2}{3}\ln(\sin(x)+2) + c$$

Home Work:

(1) $\int \sqrt{x^2 + 10x}\, dx$

(2) $\int \frac{1}{(x^2 - 6x + 34)^{\frac{3}{2}}} dx$

(3) $\int \frac{x - 1}{x^2 + x + 1} dx$

(4) $\int x(1+x)^{\frac{1}{3}} dx$

Evaluate the following integrals

1. $$\int \frac{x^2}{\sqrt{4-x^2}}\,dx$$

   $$\text{Hint : use } x = 2\sin\theta$$

2. $$\int x^3\sqrt{x^2-4}\,dx$$

   $$\text{Hint : use } x = 2\sec\theta$$

3. $$\int \sqrt{x^2+2x+2}\,dx$$

   $$\text{Hint : use } x+1 = \tan\theta$$

4. $$\int \frac{1}{\sqrt{x^2+2x+5}}\,dx$$

   $$\text{Hint : use } x+1 = 2\tan\theta$$

Evaluate the following integrals:

(1) $\int \frac{\sqrt{4-x^2}}{x^2}\,dx$

(2) $\int \frac{1}{x^2\sqrt{9+x^2}}\,dx$

(3) $\int \frac{1}{x^3\sqrt{x^2-25}}\,dx$

(4) $\int \frac{1}{(16+x^2)^{\frac{3}{2}}}\,dx$

(5) $\int \frac{\sqrt{9x^2-1}}{x}\,dx$

(6) $\int \sqrt{36-x^2}\,dx$

(7) $\int \sqrt{32-x^2-4x}\,dx$

(8) $\int \frac{1}{(x^2+4x+13)^{\frac{5}{2}}}\,dx$

(9) $\int \frac{x^2}{(1-x^2)^{\frac{3}{2}}}\,dx$

(10) $\int \frac{1}{x\sqrt{x^2-9}}\,dx$

(11) $\int \frac{1}{\sqrt{x^2-16}}\,dx$

(12) $\int \frac{1}{\sqrt{x^2+4}}\,dx$

Integrations by Complete Square

Integrations by completing square

1- Example:

Evaluate $\int \frac{2x-1}{x^2-6x+13}dx$

$$\int \frac{2x-1}{x^2-6x+13}dx$$

Solution :

$x^2-6x+13$

$1-\left(\frac{6}{2}\right)^2=9$ $\qquad$ $2-\ x^2-6x+13+9-9$

$3-\ (x^2-6x+9)+13-9$ $\qquad$ $4-\ (x-3)(x-3)+4$

$5-\ (x-3)^2+4=x^2-6x+13$

$$\int \frac{2x-1}{x^2-6x+13}dx=\int \frac{2x-1}{(x-3)^2+4}dx$$

$$u = x - 3 \Rightarrow \quad du = dx$$

$$u = x - 3 \Rightarrow \quad x = u + 3$$

$$\int \frac{2x - 1}{(x-3)^2 + 4} dx = \int \frac{2(u+3) - 1}{u^2 + 4} du$$

$$= \int \frac{2u + 6 - 1}{u^2 + 4} du = \int \frac{2u + 5}{u^2 + 4} du$$

$$= \int \frac{2u}{u^2 + 4} du + \int \frac{5}{u^2 + 4} du$$

$$= \int \frac{2u}{u^2 + 4} du + 5 \int \frac{1}{u^2 + 4} du$$

$$= \ln(u^2 + 4) + \frac{5}{2} \tan^{-1}\left(\frac{u}{4}\right) + c$$

$$\int \frac{2x - 1}{x^2 - 6x + 13} dx$$

$$= \ln((x-3)^2 + 4) + \frac{5}{2} \tan^{-1}\left(\frac{x-3}{4}\right) + c$$

2- Example:

Evaluate $\int \frac{1}{x^2+8x+25}dx$

$x^2+8x+25=?$

$1-\left(\frac{8}{2}\right)^2=16$

$2-\ x^2+8x+25+16-16$

$3-\ (x^2+8x+16)+(25-16)$

$4-\ (x+4)(x+4)+9$

$5-\ (x+4)^2+9$

$$\int \frac{1}{x^2+8x+25}dx=\int \frac{1}{(x+4)^2+9}dx$$

3- Example:

Evaluate $\int \frac{1}{(x^2+6x+13)^{\frac{3}{2}}}dx$

$$\int \frac{1}{(x^2+6x+13)^{\frac{3}{2}}}dx$$

Solution :

$$\int \frac{1}{(x^2+6x+13)^{\frac{3}{2}}}dx = \int \frac{1}{\left((x+3)^2+4\right)^{\frac{3}{2}}}dx$$

$$u = x+3 \quad \Rightarrow \quad du = dx$$

$$= \int \frac{1}{\left(u^2+4\right)^{\frac{3}{2}}}du \;\Rightarrow\; u = a\tan\theta,\; a = 2$$

$$\Rightarrow\; u = 2\tan\theta \quad \Rightarrow \quad du = 2\sec^2\theta d\theta$$

$$\int \frac{1}{\left((2\tan\theta)^2+4\right)^{\frac{3}{2}}}\cdot 2\sec^2\theta d\theta = \int \frac{1}{\left(4\tan^2\theta+4\right)^{\frac{3}{2}}}\cdot 2\sec^2\theta d\theta$$

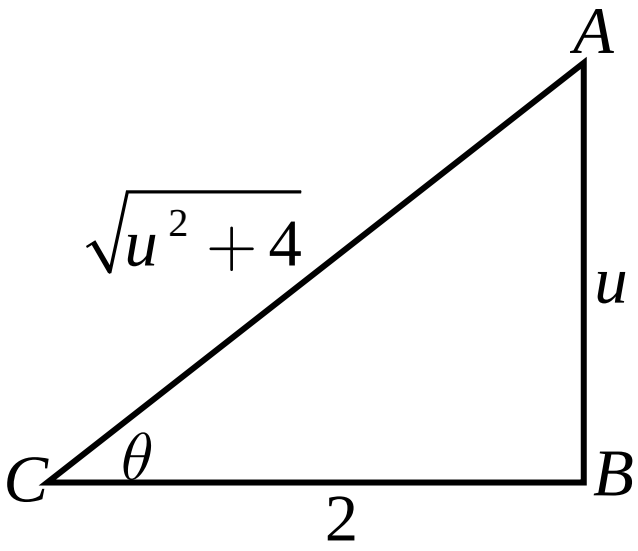
$$= \int \frac{1}{\left(4(\tan^2\theta+1)\right)^{\frac{3}{2}}}\cdot 2\sec^2\theta d\theta = \int \frac{1}{\left(4\sec^2\theta\right)^{\frac{3}{2}}}\cdot 2\sec^2\theta d\theta$$

$$= \int \frac{1}{\left(\sqrt{4\sec^2\theta}\right)^3}\cdot 2\sec^2\theta d\theta = \int \frac{1}{(2\sec\theta)^3}\cdot 2\sec^2\theta d\theta$$

$$= \int \frac{1}{8\sec^3\theta} \cdot 2\sec^2\theta \, d\theta = \int \frac{1}{4\sec\theta} \, d\theta$$

$$= \frac{1}{4}\int \frac{1}{\sec\theta} \, d\theta = \frac{1}{4}\int \cos\theta \, d\theta = \frac{1}{4}\sin\theta + c$$

$$u = 2\tan\theta \quad \Rightarrow \quad \tan\theta = \frac{u}{2}$$



$$\sin\theta = \frac{u}{\sqrt{u^2+4}}$$

$$= \frac{1}{4}\sin\theta + c = \frac{1}{4}\frac{u}{\sqrt{u^2+4}} + c$$

$$\int \frac{1}{(x^2+6x+13)^{\frac{3}{2}}}dx = \frac{1}{4}\frac{u}{\sqrt{u^2+4}} + c$$

$$= \frac{1}{4}\frac{x+3}{\sqrt{(x+3)^2+4}} + c$$

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(5) $\int \frac{3}{(x+1)\sqrt{3-x^2-2x}}dx$

Solution :

$$\sqrt{3-x^2-2x}=\sqrt{-\left(x^2+2x-3\right)}$$

$$\left(\frac{2}{2}\right)^2=1\Rightarrow\sqrt{-\left(x^2+2x-3+1-1\right)}$$

$$=\sqrt{-\left(\left(x^2+2x+1\right)-3-1\right)}$$

$$=\sqrt{-\left(\left(x^2+2x+1\right)-4\right)}$$

$$=\sqrt{-\left((x+1)(x+1)-4\right)}$$

$$\int \frac{3}{(x+1)\sqrt{3-x^2-2x}}dx$$

$$=\int \frac{3}{(x+1)\sqrt{-\left((x+1)^2-2^2\right)}}dx$$

$$\int \frac{3}{u\sqrt{-\left(u^2-2^2\right)}}du=\int \frac{3}{u\sqrt{2^2-u^2}}du$$

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$$\int \frac{3}{(x+1)\sqrt{-\left((x+1)^2-2^2\right)}}dx$$

$$\int \frac{3}{u\sqrt{-\left(u^2-2^2\right)}}du = \int \frac{3}{u\sqrt{2^2-u^2}}du$$

complete the solution

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Q: Evaluate the following integrals

(1) $\int \sqrt{32 - x^2 - 4x}\, dx$

(2) $\int \frac{1}{(x^2 + 4x + 12)^{\frac{5}{2}}} dx$

(3) $\int \frac{x + 4}{\sqrt{x^2 + 8x + 25}} dx$

(4) $\int \frac{2}{(x + 3)\sqrt{x^2 + 6x + 25}} dx$

(5) $\int \frac{3}{(x + 1)\sqrt{3 - x^2 - 2x}} dx$

(6) $\int \frac{5}{\sqrt{x^2 + 2x + 10}} dx$

(7) $\int \frac{1}{\sqrt{x^2 + 2x - 8}} dx$